APPASERVER Data Structure

September 17, 2021

SECURITY

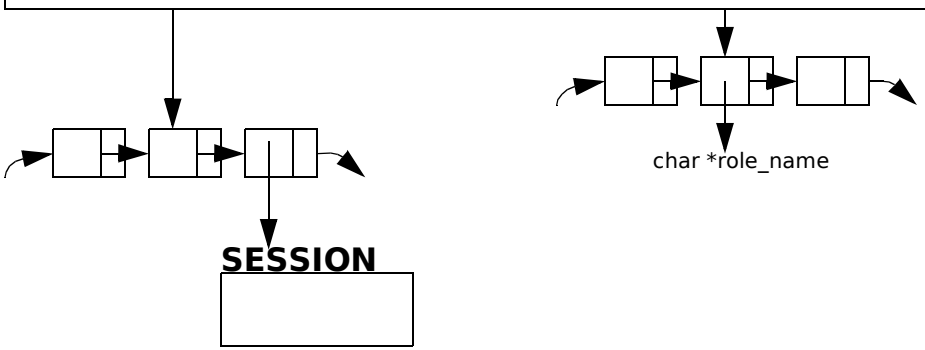
```
enum password_function password_function;{ no_encryption
                                          old_password_function
                                          regular_password_function
                                          sha2_function

enum password_function
    security_database_password_function(
        char *database_password );

char *security_encrypted_password(
    char *password_sql_injection_escape,
    enum password_function password_function );

char *security_mysql_version( void );

enum password_function
    security_mysql_version_password_function(
        security_mysql_version() );

char *security_encryption_select_clause(
    enum password_function password_function,
    char *password );

boolean security_password_match(
    database_password,
    security_encrypted_password(
        security_sql_injection_escape(
            typed_in_password ),
        security_database_password_function() );

boolean security_password_encrypted(
    char *database_password );

char *security_replace_special_characters(
    char *data );

char *security_sql_injection_escape(
    char *data )
{
    char destination[ STRING_WHERE_BUFFER ];

    return strdup(
        string_escape_character_array(
            destination,
            data,
            “‘’$;%$=_” ) );
}

boolean security_system_role(
    char *role_name );
```

SECURITY_ENTITY

```
/* Usage */
SECURITY_ENTITY *security_entity_new(
    char *login_name,
    boolean folder->non_owner_forbid,
    boolean role->override_row_restrictions );

/* Input */
char *login_name;
boolean non_owner_forbid;
boolean override_row_restrictions;

/* Process */
char *login_name_only;
char *full_name_only;
char *street_address_only;

/* Returns null if no ENTITY->login_name */
char *security_login_name_full_name_only(
    &this->street_address_only,
    this->login_name );

/* Returns full_name_only */
char *security_entity_fetch(
    &this->street_address_only,
    this->login_name );

char *security_entity_where(
    SECURITY_ENTITY *security_entity );
```

APPASERVER_USER

```
/* Usage */
APPASERVER_USER *appaserver_user_fetch(
    char *login_name,
    boolean fetch_role_name_list,
    boolean fetch_session_list );

/* Input */
char *login_name;
char *full_name;
char *typed_in_password;
char *database_password;
char *user_date_format;

/* Process */
APPASERVER_USER *appaserver_user_calloc( void );

APPASERVER_USER *appaserver_user_parse(
    char *input,
    boolean fetch_role_name_list,
    boolean fetch_session_list );

LIST *appaserver_user_role_list(
    char *login_name );

LIST *appaserver_user_session_list(
    char *login_name );

FILE *appaserver_user_update_open(
    void );

void appaserver_user_update(
    appaserver_user_update_open(),
    security_encrypted_password(
        security_sql_injection_escape(
            database_password ),
        security_mysql_version_password_function(
            security_mysql_version() ) ),
    login_name );
```



FRAMESET

```
/* Usage */
FRAMESET *frameset_new(
    char *applicatation_name,
    char *login_name,
    char *session );

/* Process */
APPASERVER_PARAMETER_FILE *
    appaserver_parameter_file =
        appaserver_parameter_file_new();

FRAMESET_LINK_FILE *frameset_link_file =
    frameset_link_file_new(
        application_name,
        appaserver_parameter_file->document_root,
        login_name,
        session );
```

FRAMESET_LINK_FILE

```
/* Usage */
FRAMESET_LINK_FILE *frameset_link_file_new(
    char *application_name,
    char *document_root_directory );

/* Process */
APPASERVER_LINK_FILE *appaserver_link_file =
    appaserver_link_file_new();

char *menu_frame_prompt_filename =
    appaserver_link_prompt_filename();

char *menu_frame_create_filename =
    appaserver_link_output_filename();

char *prompt_frame_prompt_filename =
    appaserver_link_prompt_filename();

char *prompt_frame_create_filename =
    appaserver_link_output_filename();

char *edit_frame_prompt_filename =
    appaserver_link_prompt_filename();

char *edit_frame_create_filename =
    appaserver_link_output_filename();
```

DOCUMENT

```
/* Usage */
DOCUMENT *document_new(
    char *title,
    char *role_name,
    document_head_stylesheet_string(),
    document_head_menu_setup_string(),
    document_head_calendar_setup_string(),
    document_body_onload_control_string() );

void document_quick_output(
    char *document_head_stylesheet_string() );

/* Process */
char *document_type_string(
    void );

char *document_html_standard_url(
    void );

DOCUMENT_HEAD *document_head =
    document_head_new(
        title,
        document_head_stylesheet_string(),
        document_head_menu_setup_string(),
        document_head_calendar_setup_string() );

DOCUMENT_BODY *document_body =
    document_body_new(
        role_name,
        document_body_onload_control_string() );

/* Generic */
void document_output_content_type(
    void );

void document_output_type_string(
    FILE *output_stream,
    document_type_string() );

void document_output_html_tag(
    FILE *output_stream,
    document_html_standard_url() );

void document_close(
    FILE *output_stream );
```

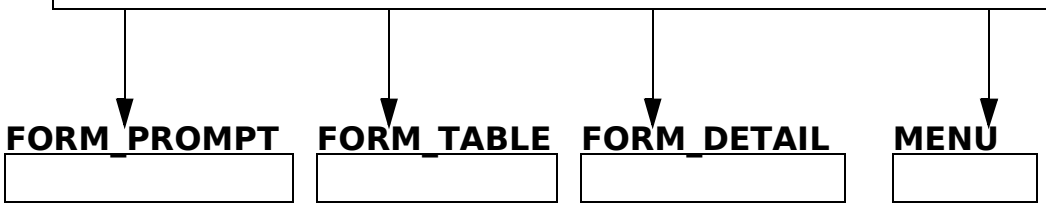
DOCUMENT_BODY

DOCUMENT_HEAD

```
/* Usage */
DOCUMENT_HEAD *document_head_new(
    char *title,
    document_head_stylesheet_string(),
    document_head_menu_setup_string(),
    document_head_calendar_setup_string() );

void document_head_quick_output(
    char *application_name );

/* Input */
char *title;
char *stylesheet_string;
char *menu_setup_string;
char *calendar_setup_string;

/* Process */
char *document_head_meta_string(
    void );

char *document_head_javascript_string(
    void );

/* Generic */
char *document_head_stylesheet_string(
    char *application_name );

char *document_head_menu_setup_string(
    void );

void document_head_output(
    FILE *output_stream,
    char *title,
    char *stylesheet_string,
    char *menu_setup_string,
    char *calendar_setup_string,
    document_head_meta_string(),
    document_head_javascript_string() );
```

DOCUMENT_BODY

```
/* Usage */
DOCUMENT_BODY *document_body_new(
    char *role_name,
    char *document_body_onload_control_string() );

/* Input */
char *onload_control_string;

MENU *menu =
    menu_fetch(
        role_name );

/* Generic */
char *document_body_onload_control_string(
    boolean with_dynarch,
    char *onload_control_string_list );

FORM_PROMPT *form_prompt_new();
FORM_TABLE *form_table_new();
FORM_DETAIL *form_detail_new();

void document_body_quick_output(
    void );
```

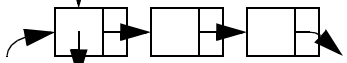


APPASERVER Data Structure

September 17, 2021

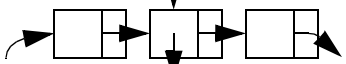
MENU

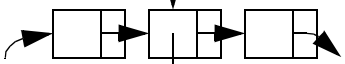
```
/* Usage */
MENU *menu_fetch(
    char *role_name );

/* Process */
char *menu_script_tag(
    void );

LIST *menu_verb_fetch_list(
    role_folder_subschema_fetch_list(),
    role_process_fetch_list(),
    role_process_set_fetch_list() );
```



MENU_VERB

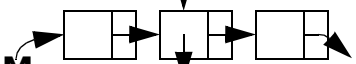
```
/* Usage */
LIST *menu_verb_fetch_list(
    role_folder_subschema_extract_list(),
    role_process_fetch_list(),
    role_process_set_fetch_list() );

/* Process */
MENU_VERB *menu_verb_subschema_fetch(
    “lookup”,
    role_folder_lookup_extract_list() );

MENU_VERB *menu_verb_subschema_fetch(
    “insert”,
    role_folder_insert_extract_list() );

LIST *menu_verb_process_group_extract_list(
    role_process_group_extract_list(),
    role_process_set_group_extract_list() );

for process_group in ‘menu_verb_process_group_extract_list()‘
{
    MENU_VERB *menu_verb_process_fetch(
        process_group,
        role_process_name_extract_list(),
        role_process_set_name_extract_list() );
}

LIST *menu_item_fetch_list(
    role_process_name_extract_list() );

LIST *menu_item_fetch_list(
    role_process_set_name_extract_list() );
```





MENU_SUBSCHEMA

```
/* Usage */
MENU_SUBSCHEMA *menu_subschema_fetch(
    char *subschema_name,
    LIST *role_folder_insert|lookup_extract_list() );

/* Process */
LIST *menu_item_fetch_list(
    role_folder_name_extract_list() );
```

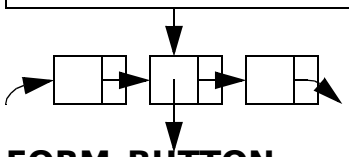


MENU_ITEM

```
/* Usage */
MENU_ITEM *menu_item_folder_new(
    char *folder_name );

MENU_ITEM *menu_item_process_new(
    char *process_name );

LIST *menu_item_fetch_list(
    role_folder_name_extract_list(
        role_folder_insert|lookup_extract_list() );

LIST *menu_item_fetch_list(
    role_process_name_extract_list() );

LIST *menu_item_fetch_list(
    role_process_set_name_extract_list() );
```

FORM_PROMPT

```
/* Usage */
FORM_PROMPT *form_prompt_calloc(
    void );

/* Attributes */
char *folder_name;
char *state;
char *application_title_string();
LIST *regular_element_list;
LIST *viewonly_element_list;
char *onload_control_string;
char *action_string;
LIST *form_button_list;
REMEMBER *remember;

/* Process */
```

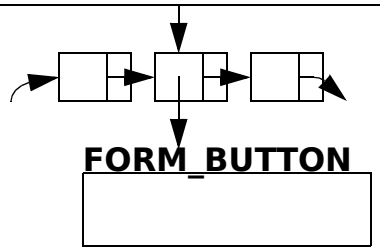


FORM_BUTTON

```
/* Usage */
FORM_BUTTON *form_button_calloc(
    void );

/* Attributes */
char *button_label;
char *onclick_control_string;

/* Process */
```

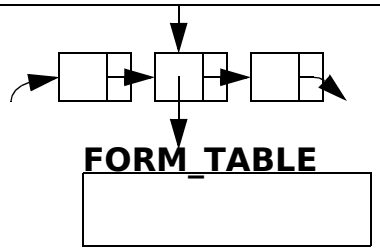
FORM_TABLE

```
/* Usage */
FORM_TABLE *form_table_calloc(
    void );

/* Attributes */
char *folder_name;
char *state;
char *application_title_string;
LIST *regular_element_list;
LIST *viewonly_element_list;
char *action_string;
LIST *form_button_list;

/* Process */
```



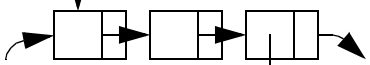
FORM_DETAIL

```
/* Usage */
FORM_DETAIL *form_detail_calloc(
    void );

/* Attributes */
LIST *form_table_list;

/* Process */
```



PROMPT_RECURSIVE

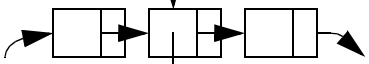
```
/* Usage */
PROMPT_RECURSIVE *prompt_recursive_new(
    char *folder_name,
    LIST *relation_mto1_non_isa_list,
    DICTIONARY *drillthru_dictionary,
    char *login_name,
    boolean drillthru_skipped );

/* Process */
if ( drillthru_skipped )
{
    return (PROMPT_RECURSIVE *)0;
}

LIST *prompt_recursive_folder_list(
    relation_mto1_non_isa_list,
    drillthru_dictionary,
    login_name );

void prompt_recursive_set_javascript(
    prompt_recursive_folder_list();

LIST *prompt_recursive_element_list(
    prompt_recursive_folder_list() );
```



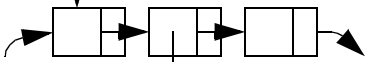
RELATION

```
/FOLDER *one_folder;
boolean prompt_mto1_recursive;
```

DICTIONARY



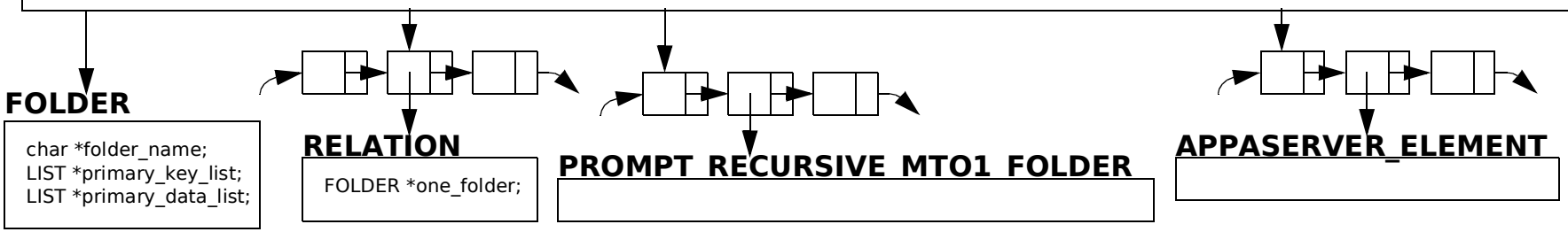
PROMPT_RECURSIVE_FOLDER



APPASERVER_ELEMENT

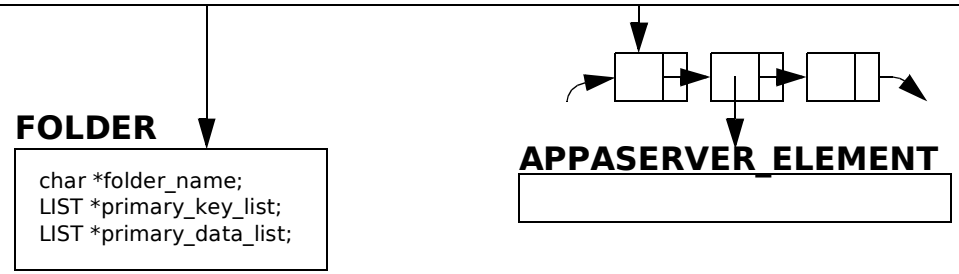
PROMPT_RECURSIVE_FOLDER

```
/* Usage */
LIST *prompt_recursive_folder_list(
    LIST *relation_mto1_non_isa_list,
    DICTIONARY *drillthru_dictionary,
    char *login_name );

PROMPT_RECURSIVE_FOLDER *
    prompt_recursive_folder_new(
        FOLDER *relation->one_folder,
        relation->drop_down_multi_select,
        DICTIONARY *drillthru_dictionary,
        char *login_name );

/* Input */
FOLDER *one_folder;
boolean drop_down_multi_select;

/* Process */
one_folder->relation_mto1_non_isa_list =
    relation_mto1_non_isa_list(
        one_folder->folder_name );

one_folder->primary_data_list =
    query_primary_delimited_fetch_list(
        one_folder->folder_name,
        one_folder->primary_key_list,
        one_folder->folder_attribute_primary_list,
        one_folder->relation_mto1_non_isa_list,
        drillthru_dictionary,
        login_name );

LIST *relation_mto1_primary_key_subset_list(
        one_folder->folder_name
            /* many_folder_name */,
        one_folder->primary_key_list );

LIST *prompt_recursive_mto1_folder_list(
    relation_mto1_primary_key_subset_list(),
    drillthru_dictionary,
    login_name );

char *prompt_recursive_folder_javascript(
    FOLDER *one_folder,
    boolean drop_down_multi_select,
    LIST *prompt_recursive_mto1_folder_list() );

LIST *prompt_recursive_folder_element_list(
    FOLDER *one_folder,
    boolean drop_down_multi_select,
    prompt_recursive_folder_javascript(),
    LIST *prompt_recursive_mto1_folder_list() );

LIST *prompt_recursive_one_folder_element_list(
    FOLDER *one_folder,
    boolean drop_down_multi_select,
    char *javascript );

LIST *prompt_recursive_mto1_folder_element_list(
    /* Returns element_list */
    prompt_recursive_one_folder_element_list(),
    prompt_recursive_folder_javascript(),
    prompt_recursive_mto1_folder_list() );
```



PROMPT_RECURSIVE_MTO1_FOLDER

```
/* Usage */
LIST *prompt_recursive_mto1_folder_list(
    LIST *relation_mto1_primary_key_subset_list(),
    DICTIONARY *drillthru_dictionary,
    char *login_name );

PROMPT_RECURSIVE_MTO1_FOLDER *
    prompt_recursive_mto1_folder_new(
        FOLDER *one_folder,
        boolean drop_down_multi_select,
        DICTIONARY *drillthru_dictionary,
        char *login_name );

/* Input */
FOLDER *one_folder;
boolean drop_down_multi_select;

/* Process */
one_folder->relation_mto1_non_isa_list =
    relation_mto1_non_isa_list(
        one_folder->folder_name );

one_folder->primary_data_list =
    query_primary_delimited_fetch_list(
        one_folder->folder_name,
        one_folder->primary_key_list,
        one_folder->folder_attribute_primary_list,
        one_folder->relation_mto1_non_isa_list,
        drillthru_dictionary,
        login_name );

/* Generic */
LIST *prompt_recursive_mto1_folder_element_list(
    /* Returns element_list */
    LIST *element_list,
    char *javascript,
    LIST *prompt_recursive_mto1_folder_list() );
```



PROMPT_EDIT_FORM

```
/* Usage */
PROMPT_EDIT_FORM *
    prompt_edit_form_fetch(
        char *application_name,
        char *login_name,
        char *session_key,
        char *folder_name,
        char *role_name,
        char *state,
        char *target_frame,
        char *appaserver_parameter_file->
            appaserver_mount_point,
        POST_DICTIONARY *
            post_dictionary_string_new() );

/* Input */
char *application_name;
char *login_name;
char *session_key;
char *folder_name;
char *role_name;
char *state;
char *target_frame
char *appaserver_mount_point;
POST_DICTIONARY *post_dictionary;

/* Process */
LIST *role_folder_list =
    role_folder_fetch_list(
        role_name,
        folder_name );

if ( boolean prompt_edit_form_forbid(
    role_folder_update( role_folder_list ),
    role_folder_lookup( role_folder_list ) )
{
    return (PROMPT_EDIT_FORM *)0;
}

ROLE *role =
    role_fetch(
        role_name,
        1 /* fetch_attribute_exclude_list */ );

FOLDER *folder =
    folder_fetch(
        folder_name,
        role_exclude_lookup_attribute_name_list(
            role->attribute_exclude_list ),
        /* Also sets folder_attribute_primary_list */
        /* and primary_key_list */
        1 /* fetch_folder_attribute_list */,
        1 /* fetch_relation_mto1_non_isa_list */,
        /* Also sets folder_attribute_append_isa_list */
        1 /* fetch_relation_mto1_isa_list */,
        1 /* fetch_relation_one2m_list */,
        1 /* fetch_relation_one2m_recursive_list */,
        0 /* not fetch_process */,
        0 /* not fetch_role_folder_list */,
        0 /* not fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

DICTIONARY_SEPARATE *dictionary_separate =
    dictionary_separate_folder_new(
        post_dictionary->original_post_dictionary,
        application_name,
        login_name,
        folder_attribute_date_name_list(
            folder->folder_attribute_isa_list ) );

DRILLTHRU *drillthru =
    drillthru_fetch(
        dictionary_separate->
            drillthru_dictionary );

PROMPT_RECURSIVE *prompt_recursive =
    prompt_recursive_new(
        folder_name,
        folder->relation_mto1_non_isa_list,
        drillthru_skipped() );

boolean prompt_edit_form_omit_insert_button(
    drillthru_skipped() );

boolean prompt_edit_form_omit_delete_button(
    list_length( folder->relation_mto1_isa_list ) );

boolean prompt_edit_form_omit_new_button(
    relation_exists_multi_select(
        folder->relation_mto1_non_isa_list ) );

folder->relation_mto1_non_isa_list =
    /* Returns relation_mto1_non_isa_list or null */
    prompt_edit_form_drillthru_skipped(
        LIST *relation_mto1_non_isa_list,
        drillthru_skipped() );

char *prompt_edit_form_target_frame(
    /* Returns target_frame or “edit_form” */
    char *target_frame,
    drillthru_skipped() );

FORM_PROMPT *form_prompt =
    form_prompt_new(
```

ROLE_FOLDER

ROLE

FOLDER

DRILLTHRU

PROMPT_RECURSIVE

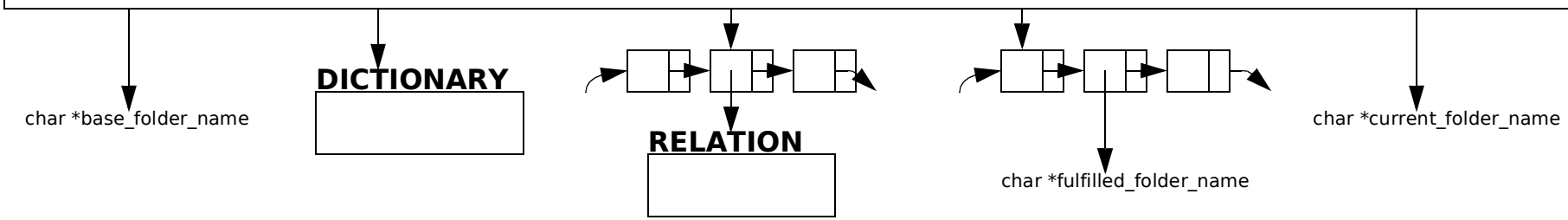
DICTIONARY_SEPARATE

```
DICTIONARY *drillthru_dictionary;
```

FORM_PROMPT

DRILLTHRU

```
/* Usage start */
DRILLTHRU *drillthru_start(
    char *base_folder_name );

void drillthru_base_folder_name_set(
    DICTIONARY *drillthru_dictionary,
    char *base_folder_name );

/* Usage */
DRILLTHRU *drillthru_fetch(
    DICTIONARY *drillthru_dictionary );

/* Input */
DICTIONARY *drillthru_dictionary;

/* Process */
DRILLTHRU *drillthru_calloc(
    void );

char *drillthru_base_folder_name(
    DICTIONARY *drillthru_dictionary );

LIST *drillthru_fulfilled_folder_name_list(
    DICTIONARY *drillthru_dictionary );

LIST *relation_mto1_drillthru_list(
    (LIST *)0 /* relation_list */,
    base_folder_name,
    drillthru_fulfilled_folder_name_list() );

char *drillthru_current_folder_name(
    relation_mto1_drillthru_list(),
    char *base_folder_name );

/* Generic */
void drillthru_fulfilled_current_folder_name_set(
    DICTIONARY *drillthru_dictionary,
    drillthru_fulfilled_folder_name_list(),
    drillthru_current_folder_name() );

void drillthru_fulfilled_folder_name_list_dictionary_set(
    DICTIONARY *drillthru_dictionary,
    drillthru_fulfilled_folder_name_list() );

boolean drillthru_skipped(
    char *folder_name,
    drillthru_base_folder_name(),
    list_length( drillthru_fulfilled_folder_name_list() ) );

boolean drillthru_middle(
    char *folder_name,
    drillthru_current_folder_name() );

boolean drillthru_finished(
    char *folder_name,
    drillthru_current_folder_name() );
```



APPASERVER Data Structure

APPASERVER_ELEMENT

```
/* Usage */
APPASERVER_ELEMENT *element_appaserver_new(
    enum element_type element_type,
    char *element_name );

/* Input */
enum element_type element_type;
char *name;

boolean remember_keystrokes_ok = 1;

/* Process */
int tab_index;
ELEMENT_DROP_DOWN *drop_down;
ELEMENT_PUSH_BUTTON *push_button;
ELEMENT_RADIO_BUTTON *radio_button;
ELEMENT_NOTEPAD *notepad;
ELEMENT_LINEBREAK *linebreak;
ELEMENT_TEXT_ITEM *text_item;
ELEMENT_PASSWORD *password;
ELEMENT_PROMPT_DATA *prompt_data;
ELEMENT_HIDDEN *hidden;
ELEMENT_UPLOAD_FILENAME *upload_filename;
ELEMENT_HTTP_FILENAME *http_filename;
ELEMENT_ANCHOR *anchor;
ELEMENT_EMPTY_COLUMN *empty_column;
ELEMENT_REFERENCE_NUMBER *reference_number;
ELEMENT_NON_EDIT_TEXT *non_edit_text;

APPASERVER_ELEMENT *
    /* Sets text_item */
    element_sort_order(
        FOLDER_ATTRIBUTE *folder_attribute );

/* Generic */
boolean element_appaserver_name_boolean(
    char *element_name );
```

drop_down	text_item	blank	table_opening
push_button	upload_filename	element_date	table_closing
radio_button	prompt_data	element_current_date	prompt
notepad	prompt_data_plus_hidden	element_time	anchor
password	reference_number	element_current_time	
non_edit_text	hidden	http_filename	
linebreak	empty_column	timestamp	

ELEMENT_DROP_DOWN

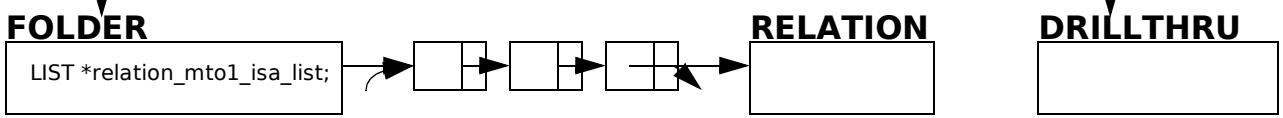
```
/* Usage */
ELEMENT_DROP_DOWN *element_drop_down_new(
    void );

/* Input */
boolean output_null_option = 1;
boolean output_not_null_option = 1;
boolean output_select_option = 1;
int date_piece_offset = -1;

/* Attributes */
char *heading;
char *initial_data;
LIST *option_data_list;
LIST *option_label_list;
int number_columns;
boolean multi_select;
char *folder_name;
char *post_change_javascript;
int max_drop_down_size;
char *multi_select_element_name;
char *onblur_javascript_function;
boolean no_initial_capital;
boolean readonly;
char *state;
int attribute_width;
```

ELEMENT_TEXT_ITEM

```
/* Usage */
ELEMENT_TEXT_ITEM *element_text_item_new(
    void );

/* Attributes */
char *data;
int attribute_width;
int widget_size;
char *heading;
char onchange_null_to_slash_yn;
char *post_change_javascript;
char *on_focus_javascript_function;
char *state;
boolean dont_create_current_date;
boolean readonly;
boolean is_numeric;
```

POST_CHOOSE_FOLDER

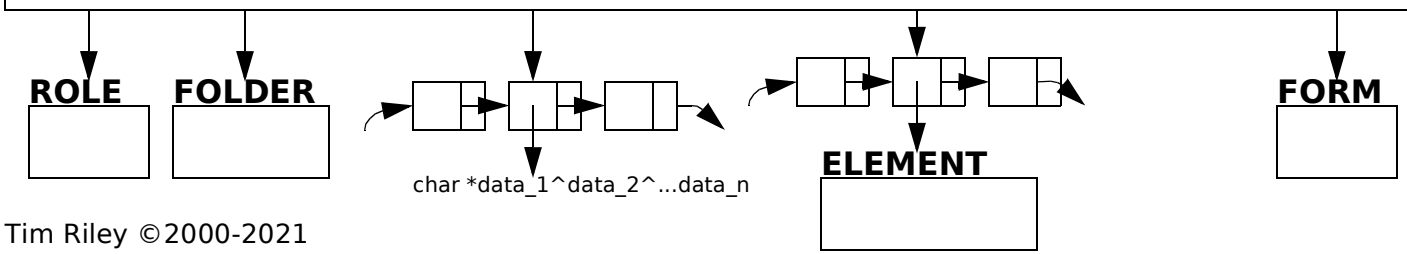
```
/* Usage */
POST_CHOOSE_FOLDER *post_choose_folder_fetch(
    /* See session_folder_integrity_exit() */
    char *application_name,
    char *login_name,
    char *session_key,
    char *role_name,
    char *folder_name,
    char *state );

/* Input */
char *application_name;
char *login_name;
char *session_key;
char *role_name;
char *folder_name;
char *state;

/* Process */
boolean post_choose_folder_fetch_relation_mto1_isa_list(
    char *state );

FOLDER *folder =
    folder_fetch(
        folder_name,
        (char *)0 /* role_name */,
        (LIST *)0 /* exclude_attribute_name_list */,
        0 /* not fetch_folder_attribute_list */,
        0 /* not fetch_relation_mto1_non_isa_list */,
        0 /* not fetch_relation_mto1_isa_list */,
        1 /* fetch_relation_one2m_list */,
        0 /* not fetch_relation_one2m_recursive_list */,
        0 /* not fetch_process */,
        0 /* not fetch_role_folder_list */,
        0 /* not fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

LIST *relation_pair_one2m_list =
    relation_pair_one2m_list(
        folder->relation_one2m_list );

DRILLTHRU *drillthru =
    drillthru_fetch(
        folder_name /* base_folder_name */,
        dictionary_small_new() /* drillthru_dictionary */ );

char *post_choose_folder_form_name(
    list_length( folder->relation_pair_one2m_list )
        /* relation_pair_one2m_list_length */,
    folder->folder_form );

boolean post_choose_folder_output_choose_isa_drop_down(
    list_length( folder->relation_mto1_isa_list )
        /* relation_mto1_isa_list_length */ );

boolean post_choose_folder_output_prompt_insert_form(
    post_choose_folder_form_name(),
    char *state );

boolean post_choose_folder_output_insert_table_form(
    post_choose_folder_form_name(),
    char *state );

boolean post_choose_folder_output_prompt_edit_form(
    post_choose_folder_form_name(),
    char *state );

boolean post_choose_folder_output_edit_table_form(
    post_choose_folder_form_name(),
    char *state );

char *post_choose_folder_system_string(
    post_choose_folder_output_choose_isa_drop_down(),
    post_choose_folder_output_prompt_insert_form(),
    post_choose_folder_output_insert_table_form(),
    post_choose_folder_output_prompt_edit_form()
    post_choose_folder_output_edit_table_form(),
    application_name,
    login_name,
    session_key,
    drillthru->current_folder_name,
    role_name,
    state,
    drillthru->drillthru_dictionary,
    list_first( folder->relation_mto1_isa_list )
        /* RELATION *first_one2m_isa_relation */ );
```



APPASERVER Data Structure

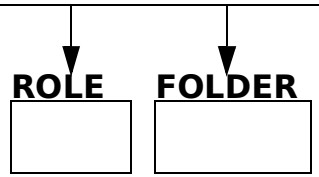
CHOOSE_ISA_DROP_DOWN (Prompt)

```
/* Usage */
CHOOSE_ISA_DROP_DOWN *choose_isa_drop_down_prompt_fetch(
    /* See session_folder_integrity_exit() */
    char *application_name,
    char *session_key,
    char *login_name,
    char *folder_name,
    char *one2m_isa_folder_name,
    char *role_name );

/* Process */
FOLDER *folder =
    folder_fetch(
        one2m_isa_folder_name,
        (char *)0 /* role_name */,
        (LIST *)0 /* exclude_attribute_name_list */,
        /* Also sets folder_attribute_primary_list */
        /* and primary_key_list */
        1 /* fetch_folder_attribute_list */,
        0 /* not fetch_relation_mto1_non_isa_list */,
        0 /* not fetch_relation_mto1_isa_list */,
        0 /* not fetch_relation_one2m_list */,
        0 /* not fetch_relation_one2m_recursive_list */,
        1 /* fetch_process */,
        0 /* not fetch_role_folder_list */,
        1 /* fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

ROLE *role =
    role_fetch(
        role_name,
        0 /* not fetch_attribute_exclude_list */ );

SECURITY_ENTITY *security_entity =
    security_entity_new(
        login_name,
        folder->non_owner_forbid,
        role->override_row_restrictions );

LIST *delimited_list =
    list_pipe_fetch(
        process_choose_isa_command_line(
            folder->populate_drop_down_process->command_line,
            application_name,
            security_entity_where(
                security_entity ),
            login_name,
            role_name,
            one2m_isa_folder_name  ) );

QUERY *query =
    query_isa_widget_new(
        one2m_isa_folder_name,
        folder->folder_attribute_primary_list,
        login_name,
        security_entity_where(
            security_entity ) );

LIST *delimited_list =
    query_delimited_list(
        query->select_clause,
        query->from_clause,
        query->where_clause,
        query->order_clause,
        0 /* max_rows */ );

LIST *choose_isa_drop_down_element_list(
    char *one2m_isa_folder_name,
    LIST *folder->primary_key_list,
    LIST *delimited_list );

char *choose_isa_drop_down_prompt_action_string(
    char *application_name,
    char *login_name,
    char *session_key,
    char *folder_name,
    char *one2m_isa_folder_name,
    char *role_name );

char *choose_isa_drop_down_prompt_message(
    char *one2m_isa_folder_name );

FORM *choose_isa_drop_down_prompt_form(
    choose_isa_drop_down_prompt_message(),
```



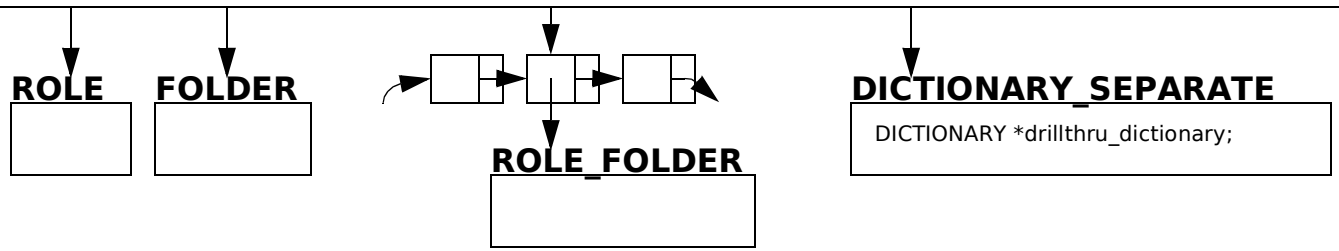
CHOOSE_ISA_DROP_DOWN (Post)

```
/* Usage */
CHOOSE_ISA_DROP_DOWN *choose_isa_drop_down_post_fetch(
    /* See session_folder_integrity_exit() */
    char *application_name,
    char *session_key,
    char *login_name,
    char *folder_name,
    char *one2m_isa_folder_name,
    char *role_name );

/* Process */
ROLE *role =
    role_fetch(
        role_name );

FOLDER *folder =
    folder_fetch(
        one2m_isa_folder_name,
        (char *)0 /* role_name */,
        (LIST *)0 /* exclude_attribute_name_list */,
        /* Also sets primary_key_list */
        1 /* fetch_folder_attribute_list */,
        0 /* not fetch_relation_mto1_non_isa_list */,
        0 /* not fetch_relation_mto1_isa_list */,
        0 /* not fetch_relation_one2m_list */,
        0 /* not fetch_relation_one2m_recursive_list */,
        1 /* fetch_process */,
        0 /* not fetch_role_folder_list */,
        1 /* fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

SECURITY_ENTITY *security_entity =
    security_entity_new(
        login_name,
        folder->non_owner_forbid,
        role->override_row_restrictions );

char *choose_isa_drop_down_post_system_string(
    char *application_name,
```



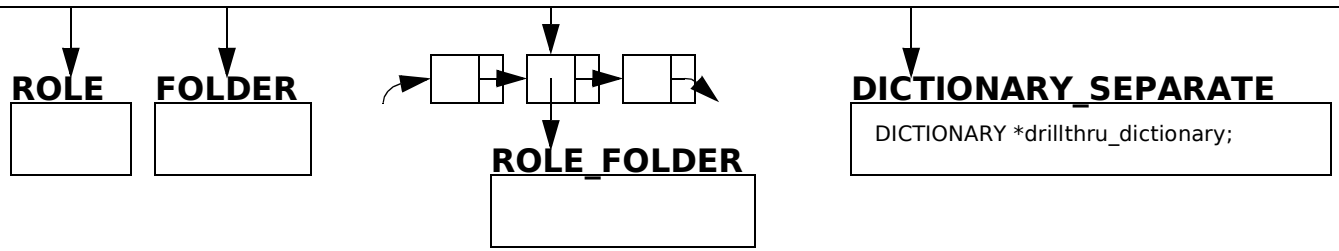
PROMPT_INSERT_FORM (Output)

```
/* Usage */
PROMPT_INSERT_FORM *
    prompt_insert_form_output_fetch(
        char *application_name,
        char *login_name,
        char *session,
        char *folder_name,
        char *role_name,
        char *state,
        char *target_frame,
        DICTIONARY_SEPARATE *dictionary_separate );

/* Process */
ROLE *role =
    role_fetch(
        role_name,
        1 /* fetch_attribute_exclude_list */ );

FOLDER *folder =
    folder_fetch(
        folder_name,
        role_exclude_insert_attribute_name_list(
            role->attribute_exclude_list ),
        /* Also sets primary_key_list */
        1 /* fetch_folder_attribute_list */,
        0 /* not fetch_relation_mto1_non_isa_list */,
        /* Also sets folder_attribute_append_isa_list */
        1 /* fetch_relation_mto1_isa_list */,
        1 /* fetch_relation_one2m_list */,
        1 /* fetch_relation_one2m_recursive_list */,
        0 /* not fetch_process */,
        0 /* not fetch_role_folder_list */,
        0 /* not fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

LIST *role_folder_list =
    role_folder_fetch_list(
        role_name,
        folder_name );

boolean prompt_insert_form_forbid(
    role_folder_insert(
        role_folder_list ) );
```



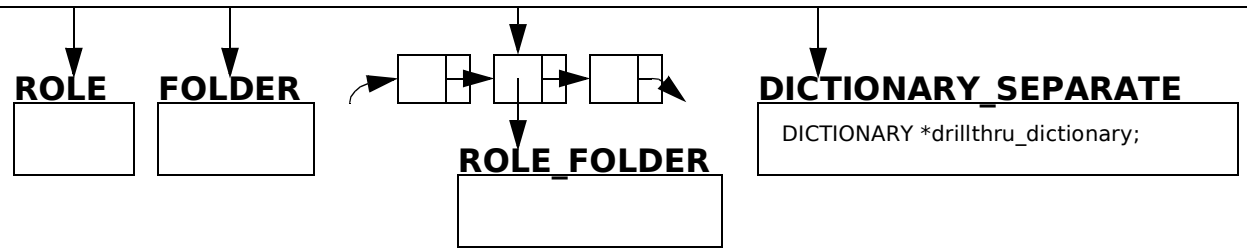
PROMPT_INSERT_FORM (Post)

```
/* Usage */
PROMPT_INSERT_FORM *
    prompt_insert_form_post_fetch(
        char *application_name,
        char *login_name,
        char *session,
        char *folder_name,
        char *role_name,
        char *state,
        char *target_frame,
        DICTIONARY_SEPARATE *dictionary_separate );

/* Process */
ROLE *role =
    role_fetch(
        role_name,
        1 /* fetch_attribute_exclude_list */ );

FOLDER *folder =
    folder_fetch(
        folder_name,
        role_exclude_insert_attribute_name_list(
            role->attribute_exclude_list ),
        /* Also sets primary_key_list */
        1 /* fetch_folder_attribute_list */,
        0 /* not fetch_relation_mto1_non_isa_list */,
        /* Also sets folder_attribute_append_isa_list */
        1 /* fetch_relation_mto1_isa_list */,
        1 /* fetch_relation_one2m_list */,
        1 /* fetch_relation_one2m_recursive_list */,
        0 /* not fetch_process */,
        0 /* not fetch_role_folder_list */,
        0 /* not fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

LIST *role_folder_list =
    role_folder_fetch_list(
        role_name,
        folder_name );

boolean prompt_insert_form_forbid(
    role_folder_insert(
        role_folder_list ) );
```



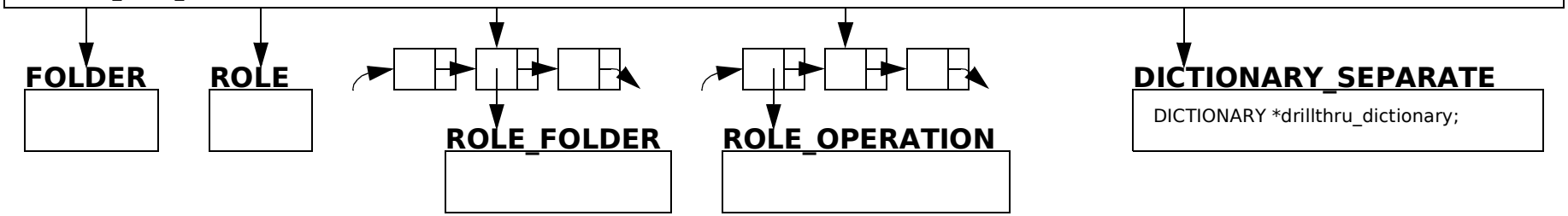
INSERT_TABLE_FORM

```
/* Usage */
INSERT_TABLE_FORM *
    insert_table_form_fetch(
        char *application_name,
        char *login_name,
        char *session_key,
        char *folder_name,
        char *role_name,
        char *target_frame,
        char *message,
        char *dictionary_string );

/* Process */
ROLE *role =
    role_fetch(
        role_name,
        1 /* fetch_attribute_exclude_list */ );

FOLDER *folder =
    folder_fetch(
        folder_name,
        role_exclude_insert_attribute_name_list(
            role->attribute_exclude_list ),
        /* Also sets primary_key_list */
        1 /* fetch_folder_attribute_list */,
        0 /* not fetch_relation_mto1_non_isa_list */,
        /* Also sets folder_attribute_append_isa_list */
        1 /* fetch_relation_mto1_isa_list */,
        1 /* fetch_relation_one2m_list */,
        1 /* fetch_relation_one2m_recursive_list */,
        0 /* not fetch_process */,
        0 /* not fetch_role_folder_list */,
        0 /* not fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

LIST *role_folder_list =
    role_folder_fetch_list(
        role_name,
        folder_name );

boolean insert_table_form_forbid(
    role_folder_insert(
        role_folder_list ) );

DICTIONARY_SEPARATE *dictionary_separate =
    dictionary_separate_string_new(
        dictionary_string );
```



EDIT_TABLE_FORM

```
/* Usage */
EDIT_TABLE_FORM *
    edit_table_form_fetch(
        char *application_name,
        char *login_name,
        char *session,
        char *folder_name,
        char *role_name,
        char *insert_update_key,
        char *target_frame,
        DICTIONARY *post_dictionary );

/* Process */
ROLE *role =
    role_fetch(
        role_name,
        1 /* fetch_attribute_exclude_list */ );

FOLDER *folder =
    folder_fetch(
        folder_name,
        role_exclude_lookup_attribute_name_list(
            role->attribute_exclude_list ),
        /* Also sets primary_key_list */
        1 /* fetch_folder_attribute_list */,
        0 /* not fetch_relation_mto1_non_isa_list */,
        /* Also sets folder_attribute_append_isa_list */
        1 /* fetch_relation_mto1_isa_list */,
        0 /* not fetch_relation_one2m_list */,
        1 /* fetch_relation_one2m_recursive_list */,
        0 /* not fetch_process */,
        0 /* not fetch_role_folder_list */,
        0 /* not fetch_row_level_restriction */,
        1 /* fetch_role_operation_list */ );

LIST *role_folder_list =
    role_folder_fetch_list(
        role_name,
        folder_name );

char *edit_table_form_state(
        role_folder_list );

DICTIONARY_SEPARATE *dictionary_separate =
    dictionary_separate_new(
        post_dictionary,
        application_name,
        folder->folder_attribute_isa_list,
        (LIST *)0 /* operation_name_list */ );

boolean edit_table_form_primary_keys_non_edit(
    list_length(
        folder->relation_mto1_isa_list ) );

LIST *folder->join_one2m_relation_list =
    relation_join_one2m_list(
        folder->relation_one2m_recursive_list,
        DICTIONARY *ignore_dictionary );

LOOKUP_BEFORE_DROP_DOWN *
    lookup_before_drop_down =
        lookup_before_drop_down_new(
```



ROW_SECURITY (1)

```
/* Usage */
ROW_SECURITY *row_security_simple_new(
    query_dictionary,
    login_name,
    full_name_only,
    street_address_only,
    folder,
    appaserver_ignore_select_attribute_name_list() );

ROW_SECURITY *row_security_detail_new(
    application_name,
    where_attribute_data_list,
    folder,
    appaserver_ignore_select_attribute_name_list() );

/* Input */
char *state;
FOLDER *folder;
ROLE *role;
char *login_name;
char *full_name_only;
char *street_address_only;
DICTIONARY *drillthru_dictionary;
DICTIONARY *query_dictionary;
DICTIONARY *sort_dictionary;
LIST *role_update_list;
FOLDER *foreign_login_name_folder;
char *attribute_not_null_string;
FOLDER *attribute_not_null_folder;
LIST *ignore_select_attribute_name_list;

/* Process */
```

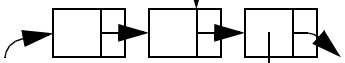
FOLDER

```
LIST *lookup_attribute_exclude_name_list;
LIST *update_attribute_exclude_name_list;
```

APPASERVER Data Structure

ROW_SECURITY (2)

```
/* Process */
LIST *row_security_role_update_list(
    void );
```

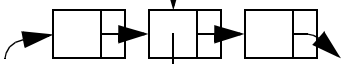


ROW_SECURITY_ROLE_UPDATE

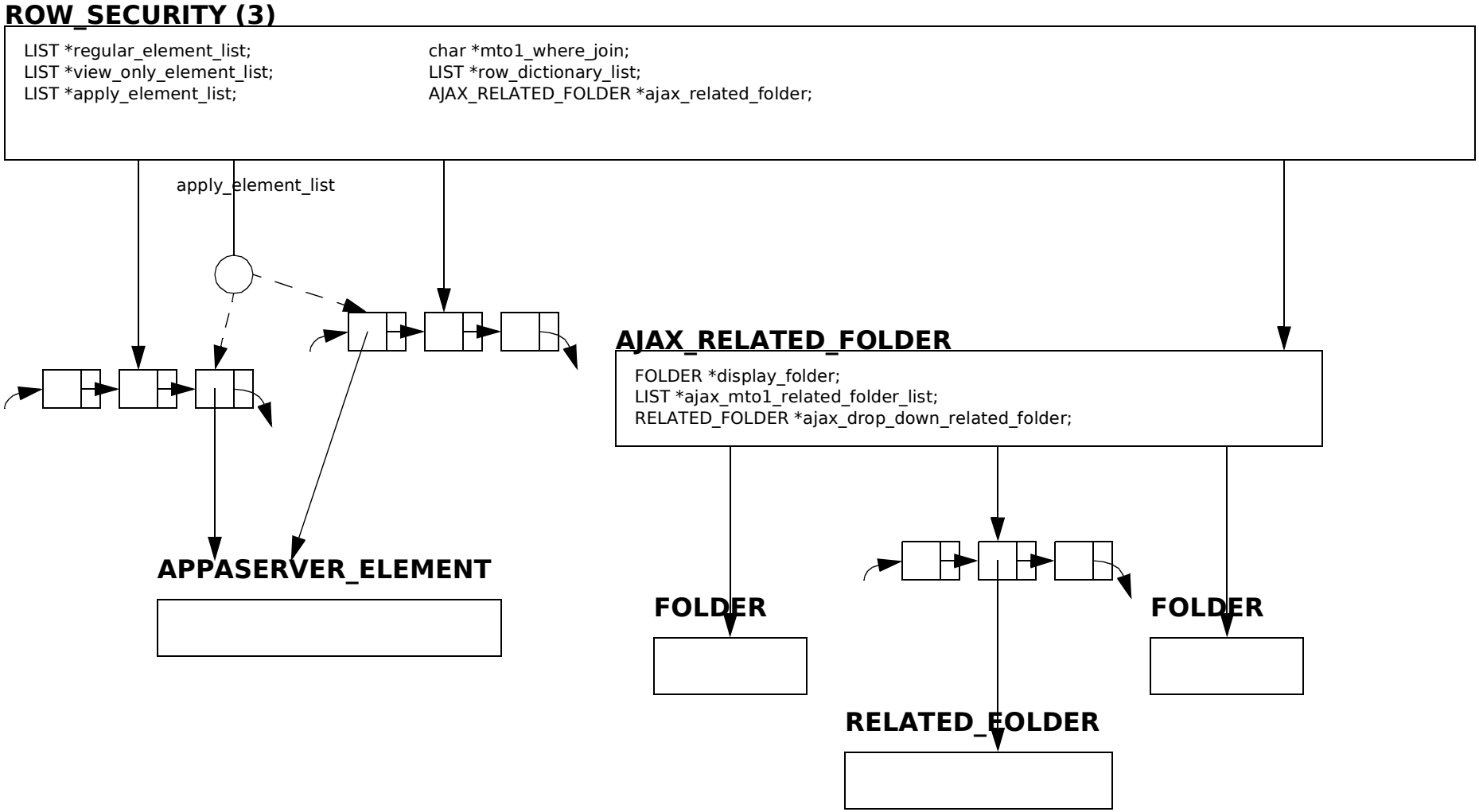
```
/* Usage */
ROW_SECURITY_ROLE_UPDATE *
    row_security_role_update_new(
        char *folder_name,
        char *attribute_not_null_string,
        ROLE *role );

FOLDER *role_update_folder;
char *attribute_not_null_string;
ROLE *role;

/* Process */
RELATED_FOLDER *row_security_role_update_related_seek(

boolean row_security_role_update_attribute_seek(
    attribute_name,
    row_security_role_update_list() );

boolean row_security_role_update_view_only(
    char *folder_name,
    FOLDER *role_update_folder,
    boolean role->override_row_restrictions,
    boolean attribute_not_null );

char *row_security_role_update_join(
```

FOLDER

```
char *folder_name;
LIST *relation_one2m_recursive_list;
```

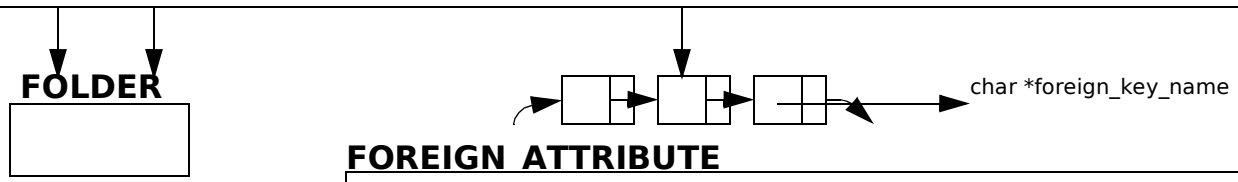


RELATION


ROW_SECURITY (3)
LIST *regular_element_list;
LIST *view_only_element_list;
LIST *apply_element_list;
char *mto1_where_join;
LIST *row_dictionary_list;
AJAX_RELATED_FOLDER *ajax_related_folder;
apply_element_list
AJAX_RELATED_FOLDER
FOLDER *display_folder;
LIST *ajax_mto1_related_folder_list;
RELATED_FOLDER *ajax_drop_down_related_folder;
APPASERVER_ELEMENT
FOLDER
FOLDER
RELATED_FOLDER

RELATION

```
/* Attributes */
char *many_folder_name;
char *one_folder_name;
char *related_attribute_name;
int pair_one2m_order;
boolean omit_one2m_detail;
boolean prompt_mto1_recursive;
boolean relation_type_isa;
boolean copy_common_attributes;
boolean automatic_preselection;
boolean drop_down_multi_select;
boolean join_one2m_each_row;
boolean ajax_fill_drop_down;
char *hint_message;

/* Process */
FOLDER *many_folder;
FOLDER *one_folder;

char *relation_system_string(
    char *where,
    char *order );

LIST *relation_system_list(
    char *relation_system_string(),
    /* Setting both will set foreign_key_list */
    boolean fetch_folder,
    boolean fetch_attribute_list );
/* Generic */
LIST *relation_one2m_pair_list(
    char *one_folder_name );

LIST *relation_mto1_non_isa_list(
    char *many_folder_name );

LIST *relation_mto1_primary_key_subset_list(
    char *many_folder_name,
    LIST *primary_key_list );

LIST *relation_one2m_list(
    char *one_folder_name );

LIST *relation_pair_one2m_list(
    relation_one2m_list() );

LIST *relation_join_one2m_list(
    relation_one2m_recursive_list(),
    DICTIONARY *ignore_dictionary );

LIST *relation_one2m_recursive_list(
    LIST *relation_list /* in/out */,
    char *one_folder_name );

LIST *relation_mto1_isa_list(
    LIST *relation_list /* in/out */,
    char *many_folder_name );

LIST *relation_mto1_drillthru_list(
    LIST *relation_list /* in/out */,
    char *base_folder_name,
    LIST *fulfilled_folder_name_list );

LIST *relation_foreign_key_list(
    folder_attribute_primary_key_list(
        one_folder->
            folder_attribute_list )
                /* primary_foreign_key_list */,
    related_attribute_name,
    foreign_attribute_name_list(
        foreign_attribute_list(
            many_folder->folder_name,
            one_folder->folder_name ) );

boolean relation_one2m_recursive_taken(
    LIST *relation_one2m_key_list_list /* in/out */,
    char *one2m_folder_name,
    LIST *relation_foreign_key_list() );

boolean consumes_taken;

RELATION *relation_consumes(
    LIST *many_attribute_name,
    relation_mto1_non_isa_list() );

boolean relation_is_primary_key_subset(
    LIST *foreign_key_list,
    LIST *many_primary_key_list );
```



FOREIGN_ATTRIBUTE

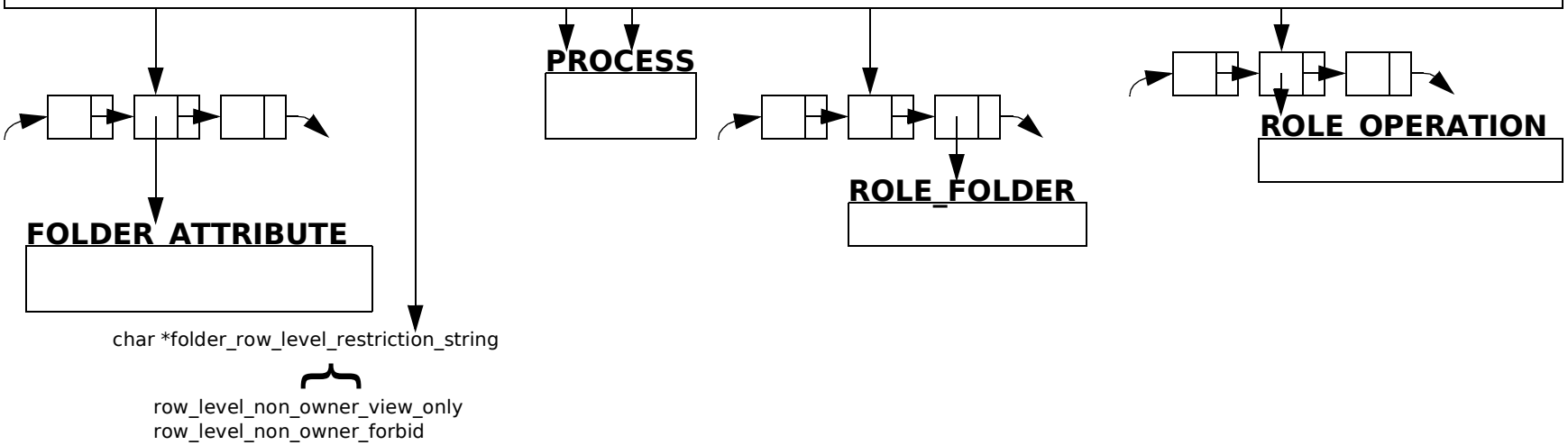
```
/* Usage */
LIST *foreign_attribute_list(
    char *many_folder_name,
    char *one_folder_name );

/* Input */
char *folder;
char *related_folder;
char *related_attribute;
char *foreign_attribute;
int foreign_key_index;

/* Process */
LIST *foreign_attribute_name_list(
    foreign_attribute_list() );
```

APPASERVER Data Structure

FOLDER (Usage)

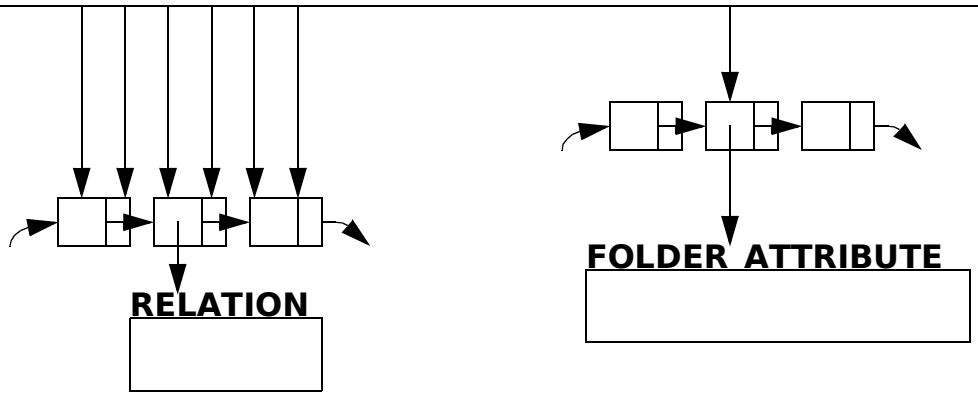
```
/* Usage */
FOLDER *folder_fetch(
    security_sql_injection_escape(
        char *folder_name ),
    /* If fetching role_folder_list */
    security_sql_injection_escape(
        char *role_name ),
    role_exclude_..._attribute_name_list(),
    /* Also sets primary_attribute_list */
    /* and primary_key_list */
    boolean fetch_folder_attribute_list,
    boolean fetch_relation_mto1_non_isa_list,
    /* Maybe sets folder_attribute_append_isa_list */
    boolean fetch_relation_mto1_isa_list,
    boolean fetch_relation_one2m_list,
    boolean fetch_relation_one2m_recursive_list,
    boolean fetch_process,
    boolean fetch_role_folder_list,
    boolean fetch_row_level_restriction,
    boolean fetch_role_operation_list );

FOLDER *folder_quick_fetch(
    char *folder_name );
```

```
/* Attributes */
char *folder_name;
char *folder_form;
int insert_rows_number;
boolean lookup_email_output;
char *notepad;
char *populate_drop_down_process_name;
char *post_change_process_name;
char *html_help_file_anchor;
char *post_change_javascript;
char *subschema;
boolean exclude_application_export;
boolean drillthru;
boolean no_initial_capital;
char *index_directory;
char *data_directory;
char *create_view_statement;
char *javascript_filename;

/* Process */
PROCESS *populate_drop_down_process;
PROCESS *post_change_process;

char *folder_primary_where(
    char *folder_name );

LIST *folder_attribute_list(
    folder_name,
    role_exclude_..._attribute_name_list() );

LIST *folder_attribute_primary_key_list(
        folder_attribute_list() );

LIST *role_folder_list =
    role_folder_fetch_list(
        sql_injection_escape_role_name,
        sql_injection_escape_folder_name ) );

LIST *role_operation_list =
    role_operation_fetch_list(
        sql_injection_escape_role_name,
        sql_injection_escape_folder_name ) );
```



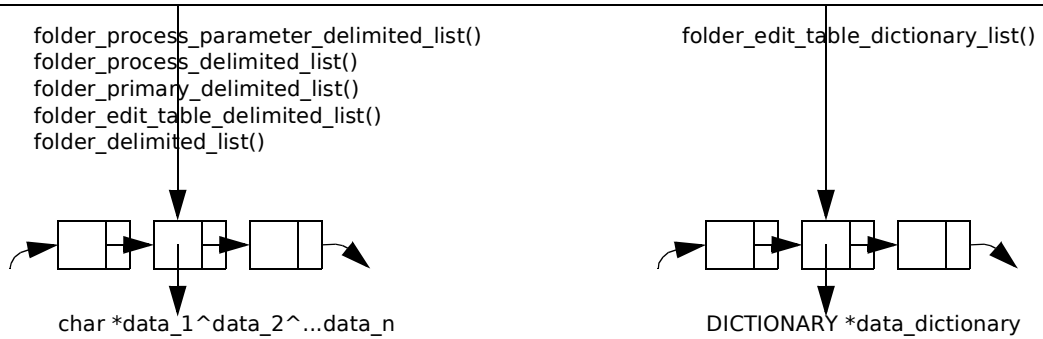
FOLDER (Relation)

```
/* Relation */
LIST *relation_mto1_isa_list(
        (LIST *)0 /* relation_list */,
        char *folder_name );

LIST *relation_mto1_non_isa_list(
    char *folder_name );

LIST *relation_one2m_list(
    folder_name );

LIST *relation_pair_one2m_list(
    relation_one2m_list() );

LIST *relation_one2m_recursive_list(
    (LIST *)0 /* one2m_recursive_list */,
    folder_name );

LIST *relation_join_one2m_list(
    relation_one2m_recursive_list() );

LIST *folder_attribute_append_isa_list(
    list_copy( folder_attribute_fetch_list() ),
    relation_mto1_isa_list() );
```



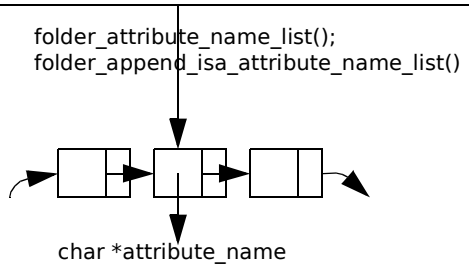
FOLDER (Data)

```
/* Data */
FOLDER *folder_process_delimited_fetch(
    char *folder_name,
    char *populate_drop_down_process_name,
    char *one2m_folder_name,
    char *state,
    char *login_name,
    char *role_name,
    DICTIONARY *drillthru_dictionary );

ROLE *role =
    role_fetch(
        role_name );

FOLDER *folder =
    folder_fetch(
        char *folder_name,
        (char *)0 /* role_name */ ),
        (LIST *)0 /* role_exclude_attribute_name_list */,
        /* Also sets primary_key_list */
        1 /* fetch_folder_attribute_list */,
        1 /* fetch_relation_mto1_non_isa_list */,
        0 /* not fetch_relation_mto1_isa_list */,
        0 /* not fetch_relation_one2m_list */,
        0 /* not fetch_relation_one2m_recursive_list */,
        0 /* not fetch_process */,
        0 /* not fetch_role_folder_list */,
        1 /* fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

SECURITY_ENTITY *security_entity =
    security_entity_new(
        login_name,
        folder->non_owner_forbid,
        role->override_row_restrictions );

LIST *folder_process_delimited_list(
    char *populate_drop_down_process_name,
    SECURITY_ENTITY *security_entity_new(),
    char *one2m_folder_name,
    char *state,
    DICTIONARY *drillthru_dictionary );

char *folder_table_name(
    char *environment_application_name(),
    char *folder_name );

LIST *folder_primary_delimited_list(
    char *folder_name,
    folder_attribute_primary_key_list(),
    folder_attribute_list(),
    relation_mto1_non_isa_list(),
    SECURITY_ENTITY *security_entity_new(),
    DICTIONARY *drillthru_dictionary,
    char *login_name );

LIST *folder_edit_table_delimited_list(
    char *folder_table_name(),
    LIST *folder_attribute_append_isa_list(),
    LIST *relation_mto1_isa_list(),
    DICTIONARY *query_dictionary );

LIST *folder_delimited_list(
    char *folder_table_name(),
    LIST *attribute_name_list,
    char *where_clause );

LIST *folder_edit_table_dictionary_list(
    char *folder_table_name(),
    LIST *folder_attribute_append_isa_list(),
    LIST *relation_mto1_isa_list(),
    DICTIONARY *query_dictionary );
```



FOLDER (Generic)

```
/* Generic */
long int folder_row_count(
    char *folder_table_name() );

char *folder_row_level_restriction_string(
    sql_injection_escape_folder_name );

boolean folder_non_owner_view_only(
    folder_row_level_restriction_string() );

boolean folder_non_owner_forbid(
    folder_row_level_restriction_string() );

boolean non_owner_view_only =
    folder_row_level_restrictions_non_owner_view_only(
        row_level_restriction_string );

boolean non_owner_forbid =
    folder_row_level_restrictions_non_owner_forbid(
        row_level_restriction_string );

LIST *folder_attribute_name_list(
    folder_attribute_list() );

LIST *folder_append_isa_attribute_name_list(
    folder_attribute_append_isa_list() );

LIST *attribute_float_list =
    folder_attribute_extract_float_list(
        folder_attribute_append_isa_list );

/* Unknown */
LIST *mto1_copy_common_related_folder_list;
LIST *copy_common_attribute_list;
LIST *append_isa_copy_folder_attribute_list;
LIST *copy_common_attribute_name_list;
LIST *append_isa_copy_attribute_name_list;
LIST *lookup_attribute_exclude_name_list;
```



FOLDER_ATTRIBUTE

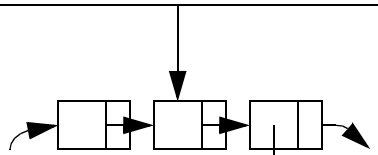
```
/* Usage */
LIST *folder_attribute_fetch_list(
    char *security_sql_injection_escape(
        folder_name ),
    LIST *exclude_attribute_name_list );

/* Attributes */
char *folder_name;
char *attribute_name;
int primary_key_index;
int display_order;
boolean omit_insert;
boolean omit_insert_prompt;
boolean omit_update;
boolean additional_unique_index;
boolean additional_index;
boolean insert_required;
boolean lookup_required;

/* Process */
ATTRIBUTE *attribute =
    attribute_fetch(
        attribute_name );

char *folder_attribute_primary_where(
    char *folder_name,
    char *attribute_name );

char *folder_attribute_where(
    char *sql_injection_escape_folder_name,
    LIST *exclude_attribute_name_list );

char *folder_attribute_system_string(
    folder_attribute_where() );

LIST *folder_attribute_system_list(
    folder_attribute_system_string() );

FOLDER_ATTRIBUTE *folder_attribute_parse(
    char *input );

/* Generic */
LIST *folder_attribute_append_isa_list(
    LIST *folder_attribute_list(),
    LIST *relation_mto1_isa_list );

LIST *folder_attribute_primary_list(
    folder_attribute_list() );

LIST *folder_attribute_primary_key_list(
    folder_attribute_list() );

LIST *folder_attribute_name_list(
    folder_attribute_list() );

LIST *folder_attribute_non_primary_key_list(
    folder_attribute_list() );

LIST *folder_attribute_date_name_list(
    folder_attribute_list() );

LIST *folder_attribute_time_name_list(
    folder_attribute_list() );

LIST *folder_attribute_date_time_name_list(
    folder_attribute_list() );

boolean folder_attribute_exists(
    char *folder_name,
    char *attribute_name );

FOLDER_ATTRIBUTE *folder_attribute_list_seek(
    char *attribute_name,
    LIST *folder_attribute_list );

char *folder_attribute_sort_attribute_name(
    LIST *folder_attribute_list );
```

ATTRIBUTE

```
/* Usage */
LIST *attribute_fetch(
    char *attribute_name );

/* Attributes */
char *attribute_name;
char *datatype_name;
int width;
int float_decimal_places;
char *hint_message;
char *post_change_javascript;
char *on_focus_javascript_function;
boolean lookup_histogram_output;
boolean lookup_time_chart_output;

/* Generic */
static LIST *attribute_list( void );

LIST *attribute_seek(
    char *attribute_name,
    LIST *attribute_list );

LIST *attribute_extract_float_list(
    LIST *attribute_list );

boolean attribute_is_date(
    char *datatype_name );

boolean attribute_is_time(
    char *datatype_name );

boolean attribute_is_number(
    char *datatype_name );

boolean attribute_is_date_time(
    char *datatype_name );

char *attribute_full_attribute_name(
    char *folder_name,
    char *attribute_name );

boolean attribute_is_yes_no(
    char *attribute_name );
```

OPERATION

```
/* Usage */
OPERATION *operation_new(
    char *operation_name,
    row_security_role_update_view_only() );

/* Input */
boolean role_update_view_only;

/* Attributes */
char *operation_name;
boolean output;

/* Process */
OPERATION *operation_fetch(
    char *operation_name );

PROCESS *process =
    process_fetch(
        operation_name,
        (char *)0 /* document_root_directory */,
        (char *)0 /* relative_source_directory */,
        1 /* check_executable_inside_filesystem */ );

boolean operation_delete_name(
    char *operation_name );

boolean operation_delete_placeholder(
    role_update_view_only,
    operation_delete_name() );

char *operation_label(
    char *operation_name,
    operation_delete_placeholder() );

/* Generic */
int operation_row_total(
    DICTIONARY *row_dictionary,
    char *operation_name,
    int highest_index );

LIST *operation_row_list(
    DICTIONARY *row_dictionary,
    char *operation_name,
    LIST *primary_key_list,
    LIST *attribute_name_list );
```



OPERATION_ROW

```
/* Usage */
OPERATION_ROW *operation_row_fetch(
    DICTIONARY *row_dictionary,
    char *operation_name,
    LIST *primary_key_list,
    LIST *attribute_name_list,
    int row_number );

/* Process */
OPERATION_ROW *operation_row_calloc(
    void );

boolean operation_row_checked(
    DICTIONARY *row_dictionary,
    char *operation_name,
    int row_number );

LIST *operation_row_primary_data_list(
    DICTIONARY *row_dictionary,
    LIST *primary_key_list,
    int row_number );

DICTIONARY *operation_single_row_dictionary(
    DICTIONARY *row_dictionary,
    LIST *attribute_name_list,
    int row_number );

/* Generic */

void operation_row_perform(
    process_operation_command_line(
        operation->process->command_line,
        environment_application_name(),
        operation_name,
        login_name,
        role_name,
        folder_name,
        getpid() /* parent_process_id */,
        session_key,
        operation_row_total(),
        operation_row_primary_data_list(),
        operation_single_row_dictionary() ) );
```

OPERATION_SEMAPHORE

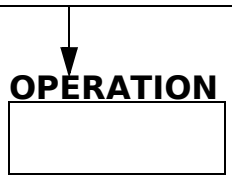
```
/* Usage */
OPERATION_SEMAPHORE *operation_semaphore_fetch(
    char *operation_name,
    char *appaserver_data_directory,
    pid_t parent_process_id,
    int operation_row_total() );

/* Process */
char *operation_semaphore_filename(
    char *operation_name,
    char *appaserver_data_directory,
    pid_t parent_process_id );

boolean operation_semaphore_group_first_time(
    operation_semaphore_filename() );

void operation_semaphore_initialize_file(
    operation_semaphore_filename() );

int operation_semaphore_row_current(
    operation_semaphore_filename() );

boolean operation_semaphore_group_last_time(
    operation_semaphore_row_current(),
    operation_row_total );

void operation_semaphore_increment(
    operation_semaphore_filename(),
    operation_semaphore_row_current() );

void operation_semaphore_remove_file(
    char *operation_semaphore_filename() );
```

SEMAPHORE

```
/* Usage */
SEMAPHORE *semaphore_new(
        key_t semaphore_key );

/* Process */
int semaphore_id(
        semaphore_key );

void semaphore_wait(
        semaphore_id() );

void semaphore_signal(
        semaphore_id() );
```

ROLE_OPERATION

```
/* Usage */
LIST *role_operation_fetch_list(
    char *folder_name,
    char *role_name );

/* Attributes */
char *folder_name;
char *role_name;
char *operation_name;

/* Process */
ROLE_OPERATION *role_operation_new(
    char *folder_name,
    char *role_name,
    char *operation_name );

char *role_operation_primary_where(
    char *folder_name,
    char *role_name,
    char *operation_name );

char *role_operation_system_string(
    char *where );

LIST *role_operation_system_list(
    role_operation_system_string() );

ROLE_OPERATION *role_operation_parse(
    char *input );

/* Generic */
OPERATION *operation =
    operation_new(
        operation_name );

LIST *role_operation_name_list(
    role_operation_fetch_list() );

OPERATION *role_operation_getset(
    LIST *operation_list,
    LIST *role_operation_list,
    char *operation_name );
```



QUERY

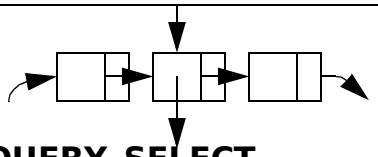
```
/* Usage */
QUERY *query_calloc(
    void );

/* Attributes */
DICTIONARY *drillthru_dictionary;
DICTIONARY *query_dictionary;
LIST *where_attribute_data_list;
char *sql_injection_escape_folder_name;
char *sql_injection_escape_role_name;
char *sql_injection_escape_login_name;
LIST *ignore_select_attribute_name_list;
char *attribute_not_null_join;
char *attribute_not_null_folder_name;
DICTIONARY *sort_dictionary;

/* Generic */
char *query_yes_no_operator_name(
    char *attribute_name );

char *query_operator_name(
    char *operator_key,
    DICTIONARY *dictionary );

enum relational_operator query_relational_operator(
    query_operator_name(),
    char *datatype_name );

char *query_sort_prefix_direction_attribute_index(
    DICTIONARY *sort_dictionary );

char *query_attribute_name_list_order(
    char *folder_name,
    LIST *primary_key_list,
    LIST *folder_attribute_append_isa_list,
    char *descending_label );

LIST *query_common_attribute_name_list();

char *query_operator_character_string(
    char *operator_string );

char *query_data_where_clause(
    char *folder_name,
    LIST *where_attribute_name_list,
    LIST *where_attribute_data_list,
    LIST *folder_attribute_list );
```

```
enum
relational_operator {
    equals
    not_equal
    not_equal_or_null
    less_than
    less_than_equal_to
    greater_than
    greater_than_equal_to
    between
    begins
    contains
    not_contains
    query_or
    is_null
    not_null
    relational_operator_empty
    relational_operator_unknown
```

APPASERVER Data Structure

QUERY (ISA WIDGET)

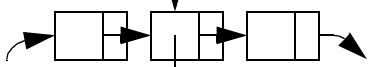
```
/* Usage */
QUERY *query_isa_widget_new(
    char *one2m_isa_folder_name,
    LIST *folder_attribute_primary_key_list(),
    LIST *folder_attribute_primary_list(),
    char *login_name,
    char *security_entity_where() );

/* Input */
char *one2m_isa_folder_name;
LIST *primary_key_list;
LIST *folder_attribute_primary_list;
char *login_name,
char *security_entity_where;

/* Process */
LIST *select_list =
    query_primary_select_list(
        primary_key_list,
        folder_attribute_primary_list,
        query_date_convert_new(
            login_name ) );

char *select_clause =
    query_select_clause(
        select_list );

char *from_clause = one2m_isa_folder_name;

char *where_clause = security_entity_where;

char *order_clause = select_clause;
```



QUERY_SELECT

QUERY (WIDGET)

```
/* Usage */
QUERY *query_widget_new(
    char *widget_folder_name,
    char *login_name,
    LIST *folder_attribute_primary_key_list(),
    LIST *folder_attribute_list(),
    LIST *relation_mto1_non_isa_list,
    char *security_entity_where(),
    DICTIONARY *drillthru_dictionary );

/* Input */
char *widget_folder_name;
char *login_name;
LIST *primary_key_list;
LIST *folder_attribute_list;
LIST *relation_mto1_non_isa_list;
char *security_entity_where;
DICTIONARY *drillthru_dictionary;

/* Process */
LIST *select_list =
    query_primary_select_list(
        primary_key_list,
        folder_attribute_list,
        query_date_convert_new(
            login_name ) );

char *select_clause =
    query_select_clause(
        select_list );

char *from_clause = widget_folder_name;

QUERY_WIDGET_WHERE *query_widget_where =
    query_widget_where_new(
        widget_folder_name,
        folder_attribute_list,
        relation_mto1_non_isa_list,
        security_entity_where,
        drillthru_dictionary );

char *where_clause =
    query_widget_where->
        where_clause;

char *order_clause = select_clause;
```

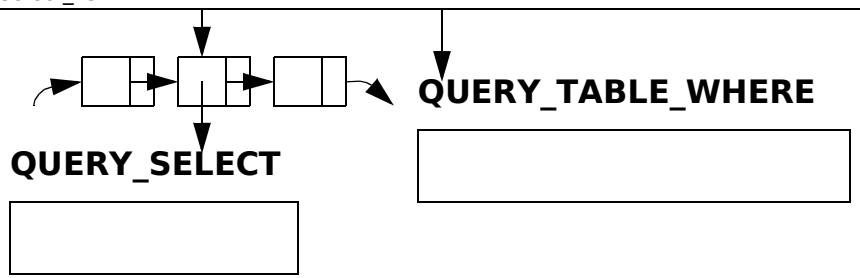


QUERY_SELECT

QUERY_WIDGET_WHERE

QUERY (TABLE)

```
/* Usage */
QUERY *query_table_new(
    char *security_sql_injection_escape(
        folder_name ),
    char *security_sql_injection_escape(
        login_name ),
    LIST *dictionary_separate->
        ignore_select_attribute_name_list,
    ROLE *role_fetch(),
    char *security_entity_where(),
    DICTIONARY *dictionary_separate->
        query_dictionary,
    DICTIONARY *dictionary_separate->
        sort_dictionary );

/* Process */
FOLDER *folder_fetch(
    security_sql_injection_escape(
        char *folder_name ),
    /* If fetching role_folder_list */
    security_sql_injection_escape(
        char *role_name ),
    role->exclude_lookup_attribute_name_list(),
    /* Also sets primary_attribute_list */
    /* and primary_key_list */
    1 /* fetch_folder_attribute_list */,
    1 /* fetch_relation_mto1_non_isa_list */,
    /* Also sets folder_attribute_append_isa_list */
    1 /* fetch_relation_mto1_isa_list */,
    0 /* not fetch_relation_one2m_list */,
    0 /* not fetch_relation_one2m_recursive_list */,
    0 /* not fetch_process */,
    1 /* fetch_role_folder_list */,
    1 /* fetch_row_level_restriction */,
    1 /* fetch_role_operation_list */ );

LIST *select_list =
    query_table_select_list(
        folder_attribute_primary_list,
        query_date_convert_new(
            login_name ) );

char *select_clause =
    query_select_clause(
        select_list );

char *from_clause =
    query_from_clause(;

QUERY_TABLE_WHERE *query_table_where =
    query_table_where_new(
        folder_attribute_list,
        relation_mto1_non_isa_list,
        security_entity_where(),
        drillthru_dictionary );

char *where_clause =
    query_table_where->
        where_clause;

char *order_clause =
    query_order_clause(
        select_clause,
        sort_dictionary );
```



QUERY (DETAIL)

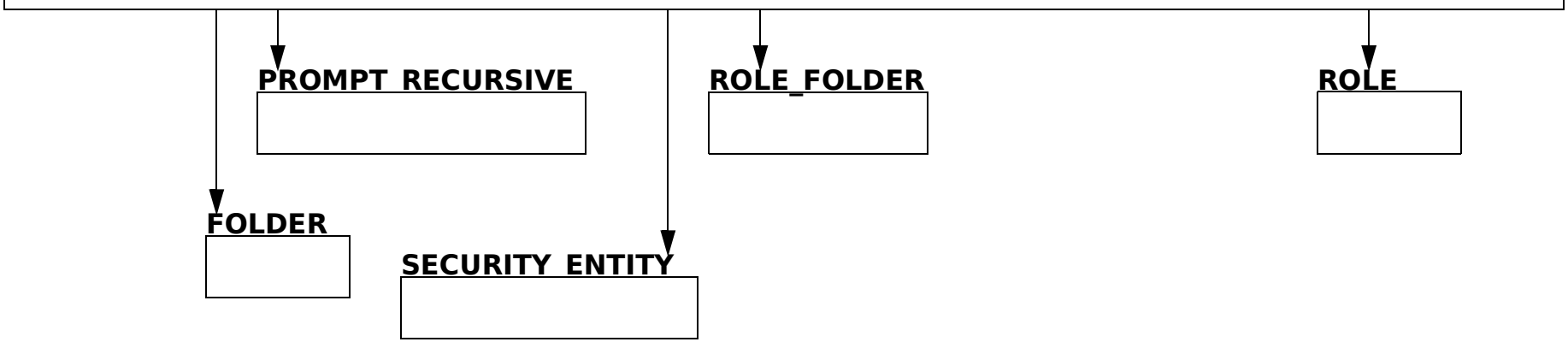
```
/* Usage */
QUERY *query_detail_new(
    LIST *where_attribute_data_list,
    char *security_sql_injection_escape(
        folder_name ),
    char *security_sql_injection_escape(
        role_name ),
    char *security_sql_injection_escape(
        login_name ),
    LIST *dictionary_separate->
        ignore_select_attribute_name_list,
    char *attribute_not_null_join,
    char *attribute_not_null_folder_name );

QUERY *query_search_replace_where_new(
    DICTIONARY *drillthru_dictionary,
    LIST *append_isa_attribute_list,
    LIST *relation_mto1_isa_list,
    boolean non_owner_forbid,
    SECURITY_ENTITY *security_entity );
```

QUERY (COMMAND_LINE_WHERE)

```
/* Usage */
QUERY *query_command_line_where_new(
    DICTIONARY *drillthru_dictionary,
    LIST *append_isa_attribute_list,
    LIST *relation_mto1_isa_list,
    boolean non_owner_forbid,
    SECURITY_ENTITY *security_entity );
```

QUERY (Process)

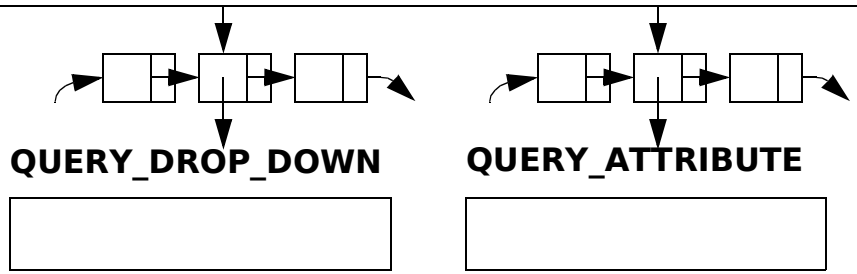
```
/* Process */
int max_rows = QUERY_MAX_ROWS;

ROLE *role =
    role_fetch(
        role_name,
        1 /* fetch_attribute_exclude_list */ );

FOLDER *query_folder =
    folder_fetch(
        sql_injection_escape_folder_name,
        sql_injection_escape_role_name,
        exclude_lookup_attribute_name_list,
        /* Also sets primary_key_list */
        1 /* fetch_folder_attribute_list */,
        0 /* not fetch_relation_mto1_non_isa_list */,
        /* Also sets folder_attribute_append_isa_list */
        1 /* fetch_relation_mto1_isa_list */,
        0 /* not fetch_relation_one2m_list */,
        0 /* not fetch_relation_one2m_recursive_list */,
        1 /* fetch_process */,
        1 /* fetch_role_folder_list */,
        1 /* fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

FOLDER *search_replace_where_folder =
    folder_fetch(
        folder_name,
        role_name,
        (LIST *)0 /* exclude_lookup_..._name_list */,
        /* Also sets primary_key_list */
        1 /* fetch_folder_attribute_list */,
        0 /* not fetch_relation_mto1_non_isa_list */,
        /* Also sets folder_attribute_append_isa_list */
        1 /* fetch_relation_mto1_isa_list */,
        0 /* not fetch_relation_one2m_list */,
        0 /* not fetch_relation_one2m_recursive_list */,
        0 /* not fetch_process */,
        0 /* not fetch_role_folder_list */,
        1 /* fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

SECURITY_ENTITY *security_entity =
    security_entity_new(
        login_name,
        query_folder->non_owner_forbid,
        role->override_row_restrictions );

PROMPT_RECURSIVE *prompt_recursive =
    prompt_recursive_new(
        query_folder,
        query_folder->mto1_related_folder_list );

ROLE_FOLDER *role_folder =
    role_folder_fetch(
        role_name,
        folder_name );
```



APPASERVER Data Structure

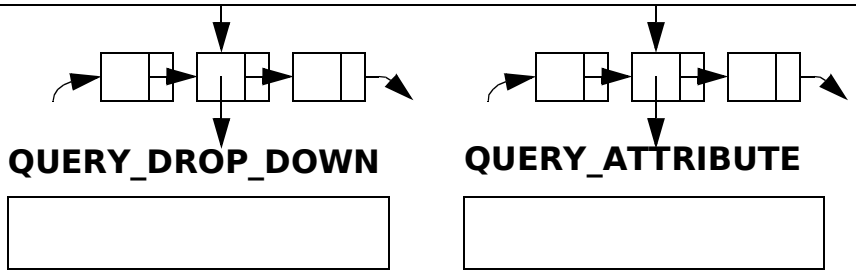
QUERY_WIDGET_WHERE

```
/* Usage */
QUERY_WIDGET_WHERE *query_widget_where_new(
    char *widget_folder_name,
    LIST *folder_attribute_list,
    LIST *relation_mto1_non_isa_list,
    char *security_entity_where(),
    DICTIONARY *drillthru_dictionary );

/* Attributes */
char *widget_folder_name;
LIST *folder_attribute_list;
LIST *primary_key_list;
LIST *relation_mto1_non_isa_list;
char *security_entity_where;
DICTIONARY *drillthru_dictionary;

/* Process */
LIST *exclude_attribute_name_list = list_new();

LIST *query_drop_down_list(
    exclude_attribute_name_list /* out */,
    widget_folder_name
        /* many_folder_name */,
    folder_attribute_list
        /* many_folder_attribute_list */,
    relation_mto1_non_isa_list,
    drillthru_dictionary );

char *query_drop_down_list_where(
    query_drop_down_list(),
    0 /* relation_mto1_isa_list_length */ );

LIST *query_attribute_list =
    query_attribute_list(
        folder_attribute_list,
        drillthru_dictionary,
        exclude_attribute_name_list );

char *query_attribute_list_where(
    query_attribute_list() );

char *where_clause =
    query_widget_where_clause(
        char *query_drop_down_where(),
        char *query_attribute_list_where(),
        char *security_entity_where() );
```



QUERY_TABLE_WHERE

```
/* Usage */
QUERY_TABLE_WHERE *query_table_where_new(
    char *folder_name,
    LIST *folder_attribute_list,
    LIST *many_primary_key_list,
    LIST *relation_mto1_non_isa_list,
    LIST *relation_mto1_isa_list,
    char *security_entity_where(),
    DICTIONARY *drillthru_dictionary );

/* Process */
LIST *exclude_attribute_name_list = list_new();

LIST *query_drop_down_list(
    exclude_attribute_name_list /* out */,
    folder_name
        /* many_folder_name */,
    folder_attribute_list
        /* many_folder_attribute_list */,
    relation_mto1_non_isa_list,
    drillthru_dictionary );

LIST *query_attribute_list =
    query_attribute_list(
        folder_attribute_list,
        drillthru_dictionary,
        exclude_attribute_name_list );

char *query_drop_down_list_where(
    query_drop_down_list(),
    list_length(
        relation_mto1_isa_list ) );

char *query_drop_down_data_where(
    char *folder_name,
    char *attribute_name,
    char *escaped_replaced_data );

char *query_attribute_list_where(
    query_attribute_list() );

char *query_widget_where_evaluate(
    query_drop_down_where(),
    query_attribute_list_where() );

LIST *query_prompt_recursive_drop_down_list(
    exclude_attribute_name_list,
    query_folder->folder_name
        /* query_folder_name */,
    prompt_recursive->
        prompt_recursive_folder_list,
    query_dictionary );

LIST *query_search_replace_where_drop_down_list(
    LIST *exclude_attribute_name_list,
    char *folder_name,
    LIST *primary_key_list,
    LIST *relation_one2m_non_recursive_list,
    LIST *relation_mto1_isa_list,
    DICTIONARY *drillthru_dictionary );

char *query_join_where(
    query_folder->folder_name,
    query_folder->primary_key_list,
    query_folder->mto1_isa_related_folder_list,
    query->attribute_not_null_join );

char *query_related_join(
    query_folder->folder_name,
    query_folder->primary_key_list,
    isa_related_folder->folder->folder_name,
    isa_related_folder->
        foreign_attribute_name_list );

char *query_primary_key_where_clause(
    LIST *primary_key_list,
    char *input,
    char delimiter );
```

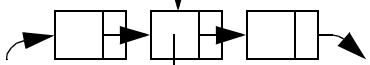


APPASERVER Data Structure

QUERY (Select)

```
LIST *select_list =
    query_select_list(
        LIST *folder_attribute_append_isa_list(),
        LIST *ignore_select_attribute_name_list,
        LIST *exclude_lookup_attribute_name_list,
        int relation_mto1_isa_length,
        query_date_convert_new() );

LIST *select_list =
    query_primary_select_list(
        LIST *folder_attribute_primary_key_list(),
        LIST *folder_attribute_primary_list(),
        query_date_convert_new() );

char *query_select_clause(
    LIST *select_list->
        query_select_string() );
```

query_select_list()
query_primary_select_list()



QUERY_SELECT

```
/* Usage */
QUERY_SELECT *query_select_new(
    char *folder_name,
    char *attribute_name,
    boolean attribute_is_date(),
    boolean attribute_is_date_time(),
    QUERY_DATE_CONVERT *
        query_date_convert );

/* Process */
char *query_select_string(
    char *folder_name,
    char *attribute_name,
    boolean attribute_is_date,
    boolean attribute_is_date_time,
    enum date_convert_format
        query_date_convert_format() );
```

QUERY_DATE_CONVERT

```
/* Usage */
QUERY_DATE_CONVERT *query_date_convert_new(
    char *login_name );

/* Process */
enum date_convert_format
    query_date_convert_format(
        environment_application_name(),
        login_name );

/* Generic */
char *query_date_convert_select_string(
    char *attribute_name,
    boolean attribute_is_date,
    boolean attribute_is_date_time,
    query_date_convert_format() );

enum date_convert_format
```

international,
american,
military,
date_convert_unknown

DATE_CONVERT

```
DATE_CONVERT *date_convert_international(
    char *application_name,
    char *login_name,
    char *date_string );

DATE_CONVERT *date_convert_string_international(
    enum date_convert_format source_format,
    char *date_string );

enum date_convert date_convert_user_date_format(
    char *application_name,
    char *login_name );
```

QUERY (From)

```
char *query_from_clause(
    char *folder_name,
    LIST *relation_mto1_isa_list,
    char *attribute_not_null_folder_name );
```

QUERY (Order)

```
char *query_order_clause(
    char *folder_name,
    LIST *primary_key_list,
    LIST *folder_attribute_append_isa_list,
    DICTIONARY *sort_dictionary );
```

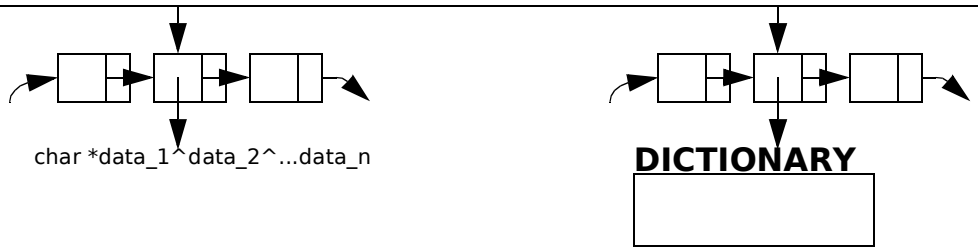
APPASERVER Data Structure

QUERY (Primary)

```
/* Usage */
LIST *query_primary_delimited_fetch_list(
    char *folder_name,
    LIST *primary_key_list,
    LIST *folder_attribute_primary_list,
    LIST *relation_mto1_non_isa_list,
    DICTIONARY *drillthru_dictionary,
    char *login_name );
```

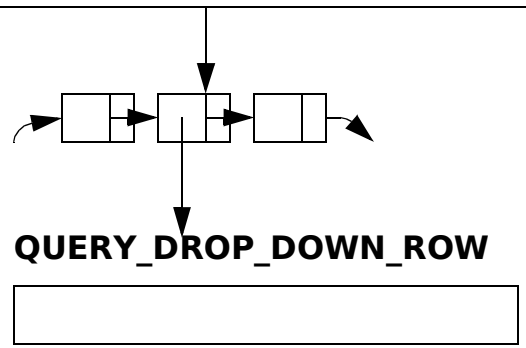
QUERY (Output)

```
/* Usage */
LIST *query_system_dictionary_list(
    query_system_string(),
    LIST *query_select_name_list(
        query->select_list ) );

LIST *query_system_delimited_list(
    query_system_string(),
    LIST *query_select_name_list(
        query->select_list ) );

/* Process */
LIST *query_dictionary_list(
    char *query->select_clause,
    LIST *query->select_list,
    char *query->from_clause,
    char *query->where_clause,
    char *query->order_clause,
    int max_rows );

LIST *query_delimited_list(
    query->select_clause,
    query->from_clause,
    query->where_clause,
    query->order_clause );

char *query_system_string(
    query->select_clause,
    query->from_clause,
    query->where_clause,
    query->order_clause,
    int max_rows );
```



QUERY_DROP_DOWN

```
/* Usage */
LIST *query_drop_down_list(
    LIST *exclude_attribute_name_list /* out */,
    char *many_folder_name,
    LIST *many_folder_attribute_list,
    LIST *relation_mto1_non_isa_list,
    DICTIONARY *dictionary_separate->drillthru_dictionary );

QUERY_DROP_DOWN *query_drop_down_new(
    char *many_folder_name,
    LIST *many_folder_attribute_list,
    char *relation->one_folder->folder_name
        /* one_folder_name */,
    LIST *relation_mto1->one_folder->primary_key_list
        /* one_primary_key_list */,
    LIST *relation_mto1->foreign_key_list,
    DICTIONARY *drillthru_dictionary,
    int dictionary_string_list_highest_index(
            relation_mto1->foreign_key_list,
            drillthru_dictionary ) );

/* Attributes */
char *many_folder_name;
char *one_folder_name;
LIST *one_primary_key_list;
LIST *foreign_key_list;
DICTIONARY *drillthru_dictionary;
int highest_index;

/* Process */
QUERY_DROP_DOWN *query_drop_down_calloc();

enum relational_operator relational_operator;

LIST *query_drop_down_row_list(
    one_primary_key_list,
    foreign_key_list,
    many_folder_attribute_list,
    drillthru_dictionary,
    highest_index );

/* Generic */
char *query_drop_down_operator_key(
    LIST *foreign_key_list );

char *query_drop_down_data_where(
    char *folder_name,
    char *attribute_name,
    char *escaped_replaced_data );
```

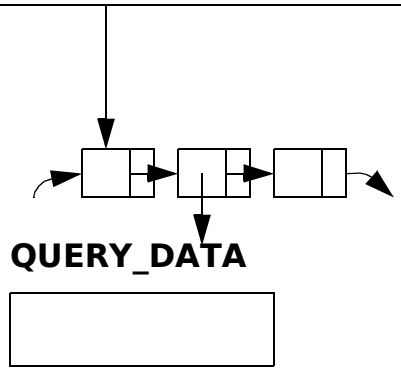


APPASERVER Data Structure

QUERY_DROP_DOWN_ROW

```
/* Usage */
LIST *query_drop_down_row_list(
    LIST *one_primary_key_list,
    LIST *foreign_key_list,
    LIST *many_folder_attribute_list,
    DICTIONARY *drillthru_dictionary,
    int dictionary_string_list_highest_index() );

QUERY_DROP_DOWN_ROW *query_drop_down_row_new(
    LIST *one_primary_key_list,
    LIST *foreign_key_list,
    LIST *many_folder_attribute_list,
    DICTIONARY *drillthru_dictionary,
    int index );

/* Input */
LIST *one_primary_key_list;
LIST *foreign_key_list;
LIST *many_folder_attribute_list;
DICTIONARY *drillthru_dictionary;
int index;
```

```
/* Process */
QUERY_DROP_DOWN_ROW *query_drop_down_row_calloc(
    void );

char *query_drop_down_row_key(
        LIST *foreign_key_list,
        int index );

char *query_drop_down_row_data_list_string(
    query_drop_down_row_key(),
    DICTIONARY *drillthru_dictionary );

LIST *query_drop_down_row_data_string_list(
        query_drop_down_row_data_list_string() );

LIST *query_data_list(
    foreign_key_list,
    many_folder_attribute_list,
    query_drop_down_row_data_string_list() );
```



QUERY_ATTRIBUTE

```
/* Usage */
LIST *query_attribute_list(
        LIST *folder_attribute_append_isa_list,
        DICTIONARY *dictionary,
        LIST *exclude_attribute_name_list );

QUERY_ATTRIBUTE *query_attribute_new(
        char *attribute_name,
        char *folder_name,
        char *datatype_name,
        enum relation_operator relation_operator;
        QUERY_DATA *from_data;
        QUERY_DATA *to_data;
        int primary_key_index );

/* Input */
char *attribute_name;
char *folder_name;
char *datatype_name;
enum relational_operator relational_operator;
QUERY_DATA *from_data;
QUERY_DATA *to_data;
int primary_key_index;

/* Process */
boolean query_attribute_date_time_between(
        QUERY_ATTRIBUTE **date_between_attribute,
        QUERY_ATTRIBUTE **time_between_attribute,
        LIST *query_attribute_list() );

char *query_attribute_between_date_time_where(
        char *date_attribute_name,
        char *time_attribute_date,
        QUERY_DATA *date_from_data,
        QUERY_DATA *time_from_data,
        QUERY_DATA *date_to_data,
        QUERY_DATA *time_to_data,
        char *folder_table_name() );

/* Generic */
char *query_attribute_operator_key(
    char *attribute_name );
```

QUERY_DATA

```
/* Usage */
LIST *query_data_list(
    LIST *foreign_key_list,
    LIST *many_folder_attribute_list,
    LIST *query_drop_down_row_data_string_list() );

QUERY_DATA *query_data_new(
    char *attribute_name,
    char *datatype_name,
    char *security_sql_injection_escape(
        security_replace_special_characters(
            string_trim_number_characters(
                char *data,
                char *datatype_name ) ) ) );

/* Input */
char *attribute_name;
char *datatype_name;
char *escaped_replaced_data;

/* Process */
char *escaped_replaced_data =
    query_data_convert_date_international(
        char *datatype_name,
        char *escaped_replaced_data );
```

QUERY_OR_SEQUENCE

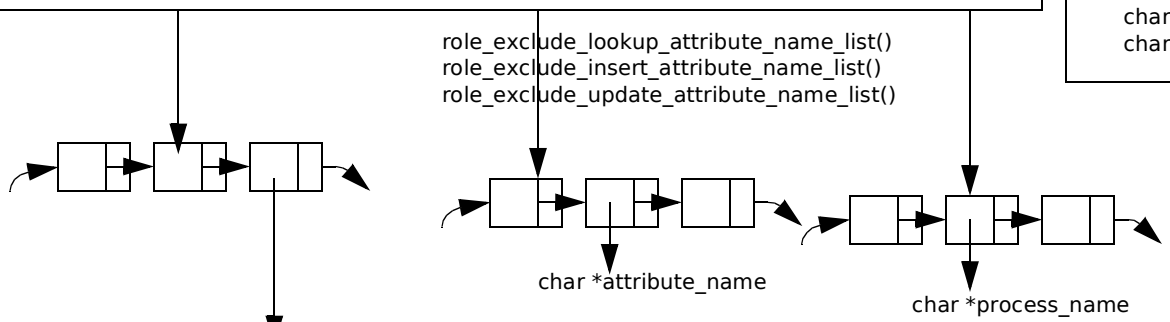
```
/* Usage */
QUERY_OR_SEQUENCE *query_or_sequence_new(
    LIST *attribute_name_list );

/* Attributes */
LIST *attribute_name_list;
LIST *data_list_list;

/* Generic */
boolean query_or_sequence_set_data_list(
    LIST *data_list_list,
    LIST *data_list );

boolean query_or_sequence_set_data_list_string(
    LIST *data_list_list,
    char *list_string );

char *query_or_sequence_where_clause(
    LIST *attribute_name_list,
    LIST *data_list_list,
    boolean with_and_prefix );
```

ROLE

```
/* Usage */
ROLE *role_fetch(
    char *role_name,
    boolean fetch_attribute_exclude_list,
    boolean fetch_process_name_list );

LIST *role_list_fetch(
    char *login_name );

/* Attributes */
char *role_name;
boolean folder_count;
boolean override_row_restrictions;

/* Process */
char *role_primary_where(
    char *role_name );

LIST *role_attribute_exclude_list =
    role_attribute_exclude_system_list();

LIST *role_exclude_lookup_attribute_name_list(
    role_attribute_exclude_system_list() );

LIST *role_exclude_insert_attribute_name_list(
    role_attribute_exclude_system_list() );

LIST *role_exclude_update_attribute_name_list(
    role_attribute_exclude_system_list() );

LIST *role_process_name_list(
    char *role_name );
```

ROLE_APPASERVER_USER

```
/* Usage */
ROLE_APPASERVER_USER *role_appaserver_user_fetch(
    char *session->login_name;
    char *security_sql_injection_escape(
        role_name ) );

LIST *role_appaserver_user_system_list(
    role_appaserver_user_system_string() );

/* Attributes */
char *login_name;
char *role_name;

/* Process */
char *role_appaserver_user_system_string(
    char *where );

char *role_appaserver_user_primary_where(
    char *login_name,
    char *role_name );
```



ROLE_ATTRIBUTE_EXCLUDE

```
/* Usage */
LIST *role_attribute_exclude_system_list(
    char *role_attribute_exclude_system_string(
        char *role_primary_where(
            char *role_name ) ) ) );

/* Attributes */
char *role_name;
char *attribute_name;
char *permission;

/* Process */
ROLE_ATTRIBUTE_EXCLUDE *
    role_attribute_exclude_parse(
        char *input );
```

insert
update
lookup

ROLE_FOLDER

```
/* Usage */
LIST *role_folder_list =
    role_folder_fetch_list(
        role_name,
        folder_name );

LIST *role_folder_subschema_fetch_list(
    char *role_name );

/* Attributes */
char *role_name;
char *folder_name;
char *permission;
char *subschema_name;

/* Process */
boolean role_folder_insert(
    role_folder_fetch_list() );

boolean role_folder_update(
    role_folder_fetch_list() );

boolean role_folder_lookup(
    role_folder_fetch_list() );

boolean role_folder_delete(
    role_operation_fetch_list() );

/* Generic */
LIST *role_folder_subschema_extract_list(
    role_folder_subschema_fetch_list() );

LIST *role_folder_lookup_extract_list(
    char *subschema_name,
    role_folder_subschema_fetch_list() );

LIST *role_folder_insert_extract_list(
    char *subschema_name,
    role_folder_subschema_fetch_list() );

LIST *role_folder_name_extract_list(
    role_folder_lookup|insert_extract_list() );
```

ROLE_PROCESS

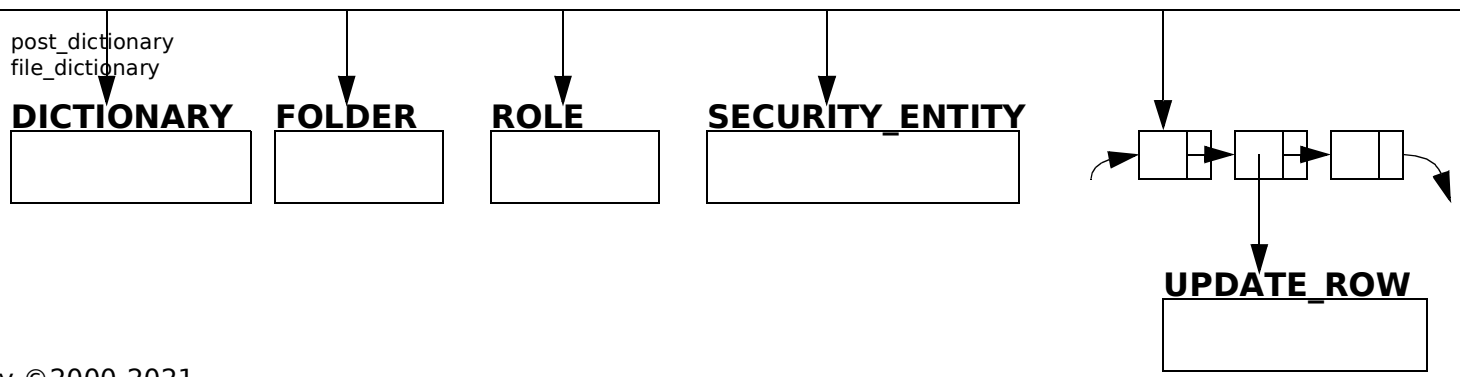
```
/* Usage */
LIST *role_process_fetch_list(
    char *role_name );

/* Attributes */
char *role_name;
char *process_name;
char *process_group_name;

/* Generic */
LIST *role_process_group_extract_list(
    role_process_fetch_list() );

LIST *role_process_name_extract_list(
    char *process_group_name,
    role_process_fetch_list() );
```

ROLE_PROCESS_SET

```
/* Usage */
LIST *role_process_set_fetch_list(
    char *role_name );

/* Attributes */
char *role_name;
char *process_set_name;
char *process_group_name;

/* Generic */
LIST *role_process_set_group_extract_list(
    role_process_set_fetch_list() );

LIST *role_process_set_name_extract_list(
    char *process_group_name,
    role_process_set_fetch_list() );
```

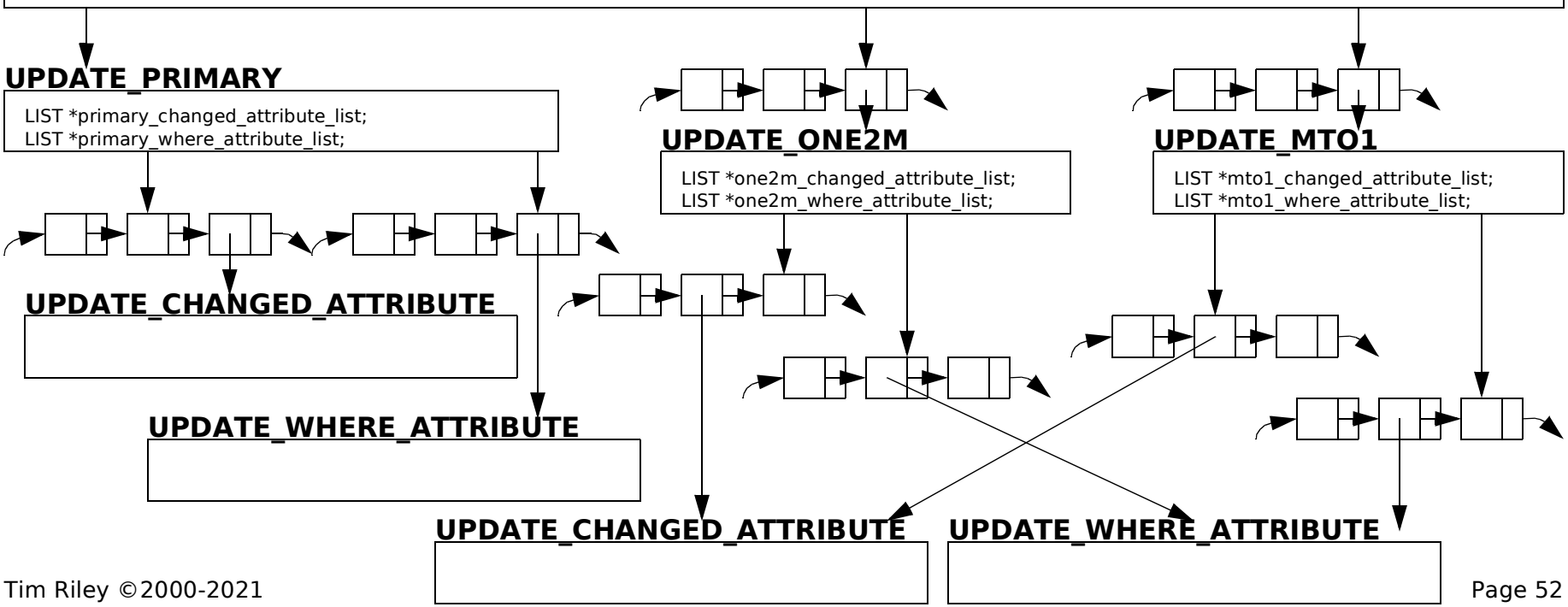
UPDATE

```
/* Usage */
UPDATE *update_new(
    /* Sets preupdate_$attribute_name for trigger */
    DICTIONARY *post_dictionary /* in/out */,
    DICTIONARY *file_dictionary,
    char *login_name,
    char *role_name,
    char *folder_name );

/* Input */
DICTIONARY *post_dictionary;
DICTIONARY *file_dictionary;
char *login_name;
char *role_name;
char *folder_name;

/* Process */
ROLE *role =
    role_fetch(
        role_name,
        1 /* fetch_attribute_exclude_list */ );

FOLDER *folder =
    folder_fetch(
        folder_name,
        role_name,
        role_exclude_update_attribute_name_list(
            role->attribute_exclude_list ),
        /* Also sets primary_key_list */
        1 /* fetch_folder_attribute_list */,
        0 /* not fetch_relation_mto1_non_isa_list */,
        /* Also sets folder_attribute_append_isa_list */
        1 /* fetch_relation_mto1_isa_list */,
        0 /* not fetch_relation_one2m_list */,
        1 /* fetch_relation_one2m_recursive_list */,
        1 /* fetch_process */,
        1 /* fetch_role_folder_list */,
        1 /* fetch_row_level_restriction */,
        0 /* not fetch_role_operation_list */ );

if ( !role_folder_update(
        folder->role_folder_list ) )
{
    return (UPDATE *)0;
}

SECURITY_ENTITY *security_entity =
    security_entity_new(
        login_name,
        folder->non_owner_forbid,
        role->override_row_restrictions );

LIST *update_row_list(
    DICTIONARY *post_dictionary,
    DICTIONARY *file_dictionary,
    FOLDER *folder,
    LIST *folder->relation_mto1_isa_list,
    LIST *folder->relation_one2m_recursive_list,
    SECURITY_ENTITY *security_entity );

char *update_set_clause(
    update_changed_attribute_list() );

char *where_clause =
    security_entity_where_clause(
        update_where_clause(
            where_attribute_list ),
        security_entity );

/* Returns mysql_message */
char *update_row_list_sql(
    char *login_name,
    SECURITY_ENTITY *security_entity,
    update_row_list() );

/* Returns mysql_message */
char *update_folder_sql(
    char *login_name,
    folder_table_name(),
    update_set_clause(),
    security_entity_where_clause() );

LIST *update_post_change_command_line_list(
    folder->post_change_process,
    folder->relation_mto1_isa_list,
    folder->relation_one2m_recursive_list,
    login_name,
    role_name,
    post_dictionary,
    update_row_list() );

/* Returns post_change_message_list_string */
char *update_post_change_execute(
    update_post_change_command_line_list() );
```



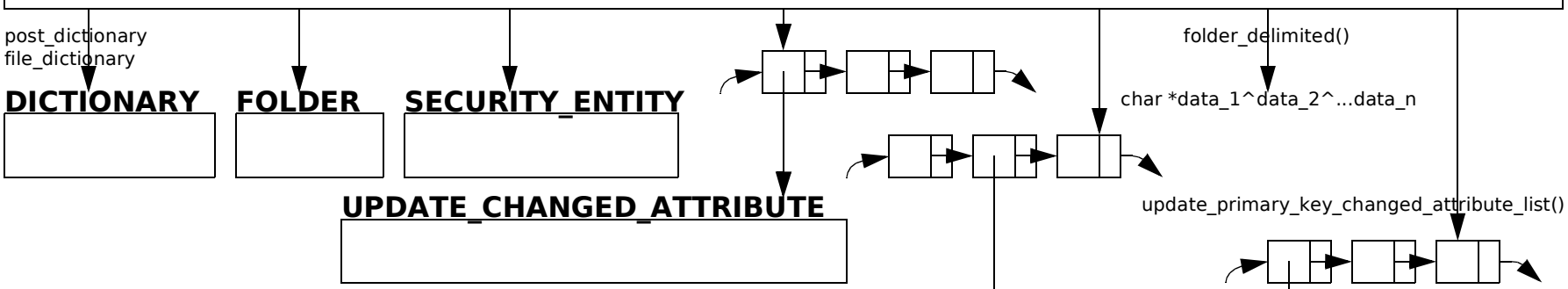
UPDATE_ROW

```
/* Usage */
UPDATE_ROW *update_row_new(
    DICTIONARY *post_dictionary,
    DICTIONARY *file_dictionary,
    FOLDER *folder,
    LIST *relation_mto1_isa_list,
    LIST *relation_one2m_recursive_list,
    SECURITY_ENTITY *security_entity,
    int row );

/* Input */
int row;

/* Process */
UPDATE_PRIMARY *update_primary =
    update_primary_new(
        post_dictionary,
        file_dictionary,
        folder,
        security_entity,
        int row );

LIST *update_row_one2m_list(
    update_primary->
        key_changed_attribute_list,
    update_primary->
        where_attribute_list,
    relation_one2m_recursive_list );

LIST *update_row_mto1_list(
    update_primary->
        key_changed_attribute_list,
    update_primary->
        where_attribute_list,
    relation_mto1_isa_list );

int update_row_cell_count(
    list_length(
        update_primary->
            changed_attribute_list )
                /* primary_cell_count */,
    update_folder_one2m_list(),
    update_folder_mto1_list() );

/* Generic */

/* Returns mysql_message */
char *update_row_sql(
    char *login_name,
    SECURITY_ENTITY *security_entity,
    UPDATE_ROW *update_row );

/* Returns return_message */
char *update_row_post_change_process(
    process_update_row_command_line(
        folder->post_change_process->command_line,
        post_dictionary,
        login_name,
        role_name,
        folder_name,
        state,
        row,
        process_name,
        one2m_folder_name,
        update_data_list(
            LIST *primary_changed_attribute_list )
            /* primary_data_list */ );
```



UPDATE_PRIMARY

```
/* Usage */
UPDATE_PRIMARY *update_primary_new(
    /* Sets preupdate_$attribute_name for trigger */
    DICTIONARY *post_dictionary,
    DICTIONARY *file_dictionary,
    FOLDER *folder,
    SECURITY_ENTITY *security_entity,
    int row );

/* Process */
LIST *update_primary_changed_attribute_list(
    /* Sets preupdate_$attribute_name for trigger */
    post_dictionary,
    file_dictionary,
    folder->folder_attribute_list,
    row );

LIST *update_primary_where_attribute_list(
    file_dictionary,
    folder->primary_key_list,
    row );

char *primary_where_clause =
    security_entity_where_clause(
        update_where_clause(
            update_primary_where_attribute_list(),
        security_entity );

char *folder_delimited(
    folder_table_name(
        environment_application_name(),
        folder->folder_name ),
    folder->primary_key_list,
    primary_where_clause );

/* Generic */
LIST *update_primary_key_changed_attribute_list(
    update_primary_changed_attribute_list() );
```



UPDATE_WHERE_ATTRIBUTE

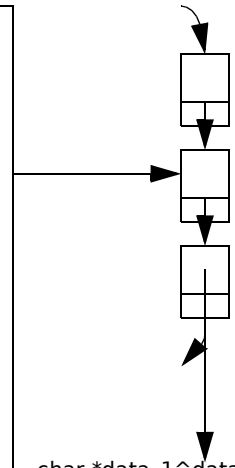
```
/* Usage */
UPDATE_WHERE_ATTRIBUTE *
    update_where_attribute_new(
        DICTIONARY *file_dictionary,
        char *primary_key,
        int row );

LIST *update_foreign_where_attribute_list(
        LIST *primary_key_changed_attribute_list,
        LIST *foreign_key_list );

/* Process */
char *attribute_name;
char *data;

char *update_where_clause(
    LIST *where_attribute_list );
```

UPDATE_CHANGED_ATTRIBUTE

APPASERVER Data Structure

UPDATE_ONE2M

```
/* Usage */
UPDATE_ONE2M *update_one2m_new(
    LIST *primary_key_changed_attribute_list,
    RELATION *relation_one2m );

/* Input */
LIST *primary_key_changed_attribute_list;
RELATION *relation_one2m;

/* Process */
LIST *update_foreign_changed_attribute_list(
    LIST *primary_key_changed_attribute_list,
    char *relation_one2m->many_folder->folder_name,
    LIST *relation_one2m->foreign_key_list );

LIST *update_foreign_where_attribute_list(
    LIST *primary_key_changed_attribute_list,
    LIST *relation_one2m->foreign_key_list );

LIST *folder_delimited_list(
    folder_table_name(
        environment_application_name(),
        relation_one2m->
            many_folder->
            folder_name )
                /* table_name */,
    LIST *relation_one2m->
        many_folder->
        primary_key_list
            /* attribute_name_list */,
    update_where_clause(
        update_foreign_where_attribute_list() )
            /* where_clause */ );
```



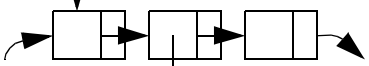
char *data_1^data_2^...data_n

primary_key_changed_attribute_list

RELATION *relation_one2m

update_foreign_changed_attribute_list()

update_foreign_where_attribute_list()

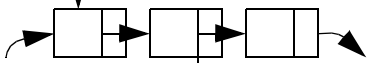


UPDATE_CHANGED_ATTRIBUTE

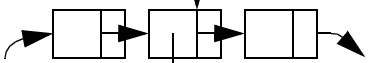
```
FOLDER_ATTRIBUTE *folder_attribute;
UPDATE_CHANGED_ATTRIBUTE_DATA *
        update_changed_attribute_data;
```

RELATION

```
FOLDER *many_folder;
LIST *foreign_key_list;
```



UPDATE_WHERE_ATTRIBUTE

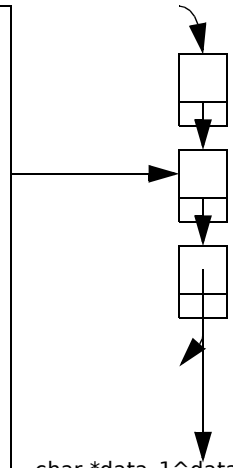


UPDATE_CHANGED_ATTRIBUTE

APPASERVER Data Structure

UPDATE_MTO1

```
/* Usage */
UPDATE_MTO1 *update_mto1_new(
    LIST *primary_key_changed_attribute_list,
    RELATION *relation_mto1 );

/* Input */
LIST *primary_key_changed_attribute_list;
RELATION *relation_mto1;

/* Process */
LIST *update_foreign_changed_attribute_list(
    LIST *primary_key_changed_attribute_list,
    char *relation_mto1->one_folder->folder_name,
    LIST *relation_mto1->foreign_key_list );

LIST *update_foreign_where_attribute_list(
    LIST *primary_key_changed_attribute_list,
    LIST *relation_mto1->foreign_key_list );

LIST *folder_delimited_list(
    folder_table_name(
        environment_application_name(),
        relation_mto1->
            one_folder->
            folder_name )
                /* table_name */,
    LIST *relation_mto1->
        one_folder->
        primary_key_list
                /* attribute_name_list */,
    update_where_clause(
        update_foreign_where_attribute_list() )
                /* where_clause */ );
```



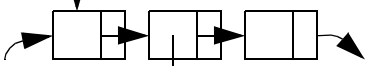
char *data_1^data_2^...data_n

primary_key_changed_attribute_list

RELATION *relation_one2m

update_foreign_changed_attribute_list()

update_foreign_where_attribute_list()

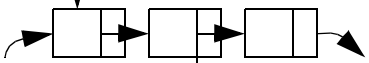


UPDATE_CHANGED_ATTRIBUTE

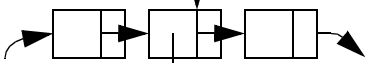
```
FOLDER_ATTRIBUTE *folder_attribute;
UPDATE_CHANGED_ATTRIBUTE_DATA *
        update_changed_attribute_data;
```

RELATION

```
FOLDER *one_folder;
LIST *foreign_key_list;
```



UPDATE_WHERE_ATTRIBUTE



UPDATE_CHANGED_ATTRIBUTE

UPDATE_CHANGED_ATTRIBUTE

```
/* Usage */
LIST *update_changed_attribute_list(
    DICTIONARY *post_dictionary,
    DICTIONARY *file_dictionary,
    LIST *folder_attribute_list,
    int row );

UPDATE_CHANGED_ATTRIBUTE *update_changed_attribute_fetch(
    /* Sets preupdate_$attribute_name for trigger */
    DICTIONARY *post_dictionary,
    DICTIONARY *file_dictionary,
    FOLDER_ATTRIBUTE *folder_attribute,
    int row );

UPDATE_CHANGED_ATTRIBUTE *update_changed_attribute_new(
    char *folder_name,
    char *attribute_name );

/* Input */
FOLDER_ATTRIBUTE *folder_attribute;
```

```
/* Process */
UPDATE_CHANGED_ATTRIBUTE_DATA *update_changed_attribute_data(
    post_dictionary,
    file_dictionary,
    folder_attribute->attribute->attribute_name,
    folder_attribute->attribute->datatype_name,
    folder_attribute->primary_key_index,
    row );

char *update_changed_attribute_preupdate_label(
    char *attribute_name,
    int row );

dictionary_set(
    post_dictionary,
    update_changed_attribute_preupdate_label,
    update_changed_attribute_data->old_data );

LIST *update_primary_changed_attribute_list(
    update_changed_attribute_list() );
```

FOLDER_ATTRIBUTE

```
FOLDER *folder;
ATTRIBUTE *attribute;
int primary_key_index;
```

FOLDER

```
char *folder_name;
```

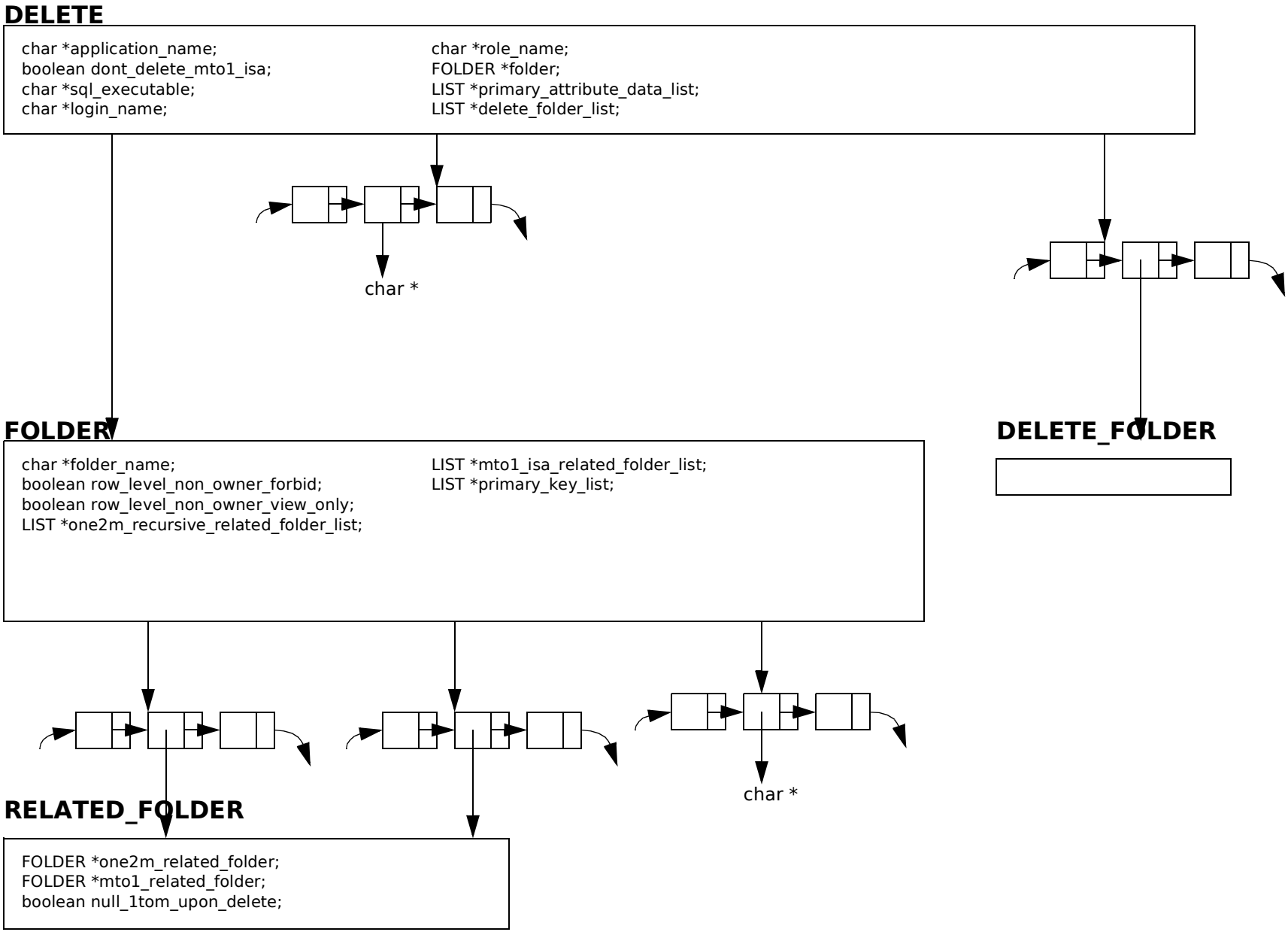
ATTRIBUTE

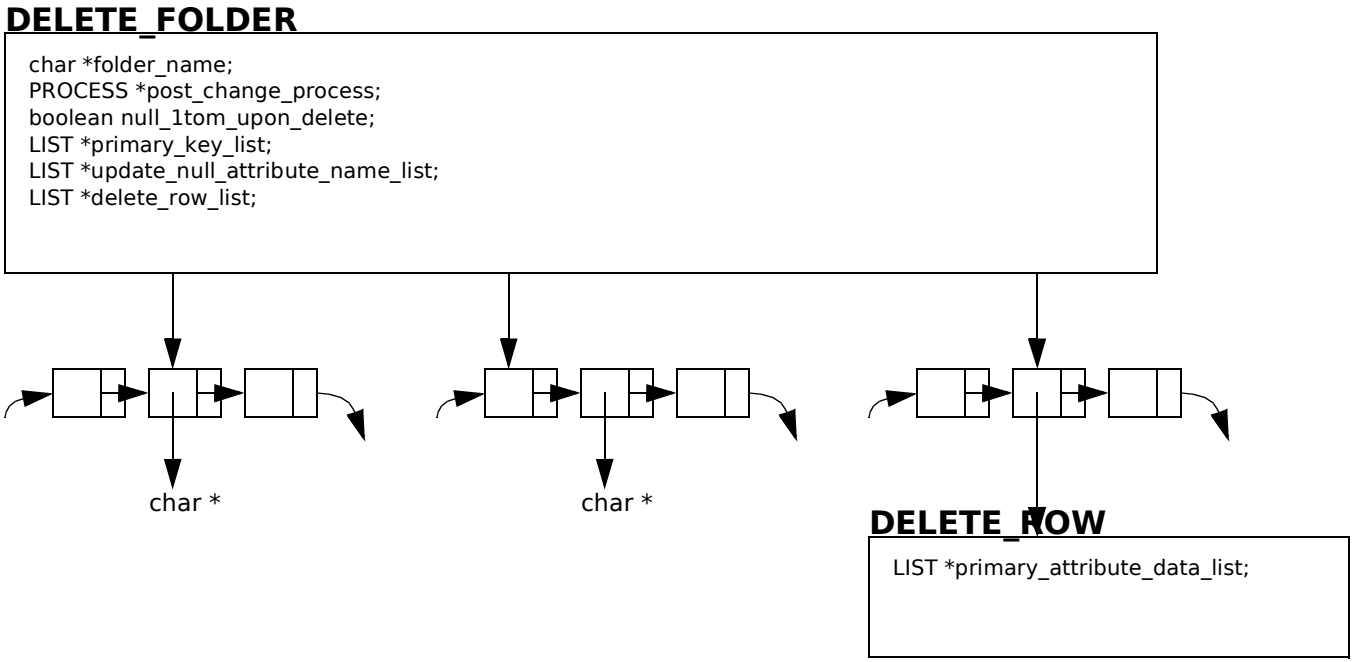
```
char *attribute_name;
char *datatype_name;
```

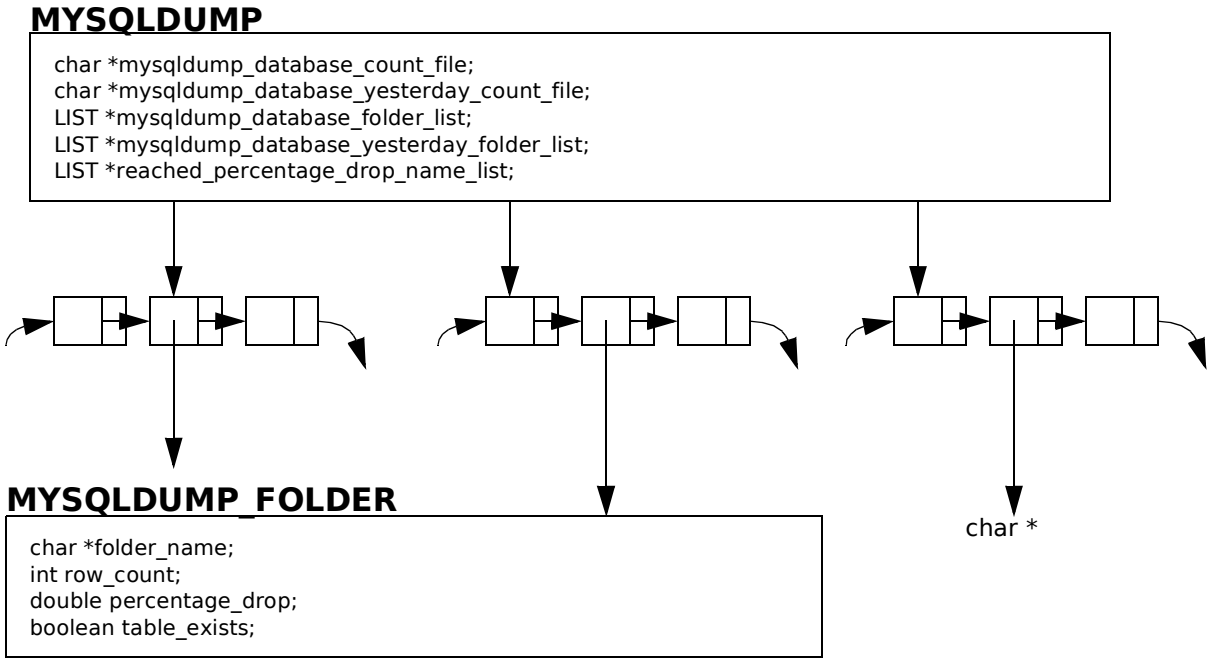
UPDATE_CHANGED_ATTRIBUTE_DATA

```
/* Usage */
UPDATE_CHANGED_ATTRIBUTE_DATA *
    update_changed_attribute_data_fetch(
        DICTIONARY *post_dictionary,
        DICTIONARY *file_dictionary,
        char *attribute_name,
        char *datatype_name,
        int primary_key_index,
        int row );

UPDATE_CHANGED_ATTRIBUTE_DATA *
    update_changed_attribute_data_new(
        char *key,
        char *old_data,
        char *escaped_replaced_new_data );

/* Process */
char *key =
    update_dictionary_key(
        attribute_name,
        row );
```

```
char *old_data =
    dictionary_get(
        key,
        file_dictionary );

char *new_data =
    dictionary_get(
        key,
        post_dictionary );

boolean changed_attribute =
    ( strcasecmp( old_data, new_data ) != 0 );

char *escaped_replaced_new_data =
        security_sql_injection_escape(
            security_replace_special_characters(
                string_trim_number_characters(
                    new_data,
                    datatype_name ) ) );
```


DELETE
char *application_name;
boolean dont_delete_mto1_isa;
char *sql_executable;
char *login_name;
char *role_name;
FOLDER *folder;
LIST *primary_attribute_data_list;
LIST *delete_folder_list;
char *
DELETE_FOLDER
FOLDER
char *folder_name;
boolean row_level_non_owner_forbid;
boolean row_level_non_owner_view_only;
LIST *one2m_recursive_related_folder_list;
LIST *mto1_isa_related_folder_list;
LIST *primary_key_list;
char *
RELATED_FOLDER
FOLDER *one2m_related_folder;
FOLDER *mto1_related_folder;
boolean null_1tom_upon_delete;


DELETE_FOLDER
char *folder_name;
PROCESS *post_change_process;
boolean null_1tom_upon_delete;
LIST *primary_key_list;
LIST *update_null_attribute_name_list;
LIST *delete_row_list;
char *
char *
DELETE_ROW
LIST *primary_attribute_data_list;

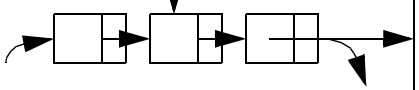


STRING

```
boolean string_strcpy(
    char *destination,
    char *source,
    int buffer_size );

boolean string_strcmp(
    char *string1,
    char *string2 );

char *string_escape_character(
    char *destination,
    char *data,
    char character );

char *string_escape_character_array(
    char *destination,
    char *data,
    char *character_array );

/* Position is 1 based */
int string_character_position(
    char *data,
    char character );

char *string_trim_character(
    char *data,
    char character );

char *string_trim_character_array(
    char *data,
    char *character_array );

char *string_trim_number_characters(
    char *number,
    char *datatype_name )
{
    return string_trim_character_array(
        number,
        ",$" );
}

char *string_decode_html_post(
    char *destination,
    char *source );

char *string_trim(
    char *buffer );

char *string_extract_lt_gt_delimited(
    char *destination,
    char *source );

int string_index(
    char *source );
```

PAIR_ONE2M

```
char *many_folder_name;
char *one_folder_name;

PAIR_ONE2M *pair_one2m_prompt_form_new(
    char *one_folder_name,
    char *keystrokes_save_function );

LIST *pair_one2m_prompt_form_folder_list(
    char *keystrokes_save_function,
    LIST *relation_one2m_pair_relation_list() );

char *pair_one2m_prompt_element_name(
    char *PAIR_ONE2M_PREFIX,
    char *folder_label );

boolean pair_one2m_participating(
    DICTIONARY *pair_one2m_dictionary );

PAIR_ONE2M *pair_one2m_post_new(
    DICTIONARY *pair_one2m_dictionary );

boolean pair_one2m_duplicate(
    char *MYSQL_DUPLICATE_ERROR_MESSAGE_KEY,
    char *message );

void pair_one2m_duplicate_set(
    DICTIONARY *pair_one2m_dictionary,
    char *PAIR_ONE2M_DUPLICATE_KEY );

boolean pair_one2m_inserted_duplicate(
    char *PAIR_ONE2M_DUPLICATE_KEY,
    DICTIONARY *pair_one2m_dictionary );

LIST *pair_one2m_fulfilled_folder_name_list(
    char *PAIR_ONE2M_FULFILLED_LIST_LABEL,
    DICTIONARY *pair_one2m_dictionary );

char *pair_one2m_folder_name(
    char *folder_label,
    DICTIONARY *pair_one2m_dictionary );

void pair_one2m_folder_set(
    DICTIONARY *pair_one2m_dictionary,
    char *folder_label,
    char *folder_name );

char *pair_one2m_next_folder_name(
    LIST *pair_one2m_fulfilled_folder_name_list(),
    LIST *relation_one2m_pair_relation_list(),
    char *one_folder_name,
    char *many_folder_name );

DICTIONARY *pair_one2m_fulfilled_dictionary(
    DICTIONARY *pair_one2m_dictionary,
    char *PAIR_ONE2M_FULFILLED_LIST_LABEL,
    LIST *pair_one2m_fulfilled_folder_name_list );

/* Insert form */
PAIR_ONE2M *pair_one2m_insert_form_new(
    DICTIONARY *pair_one2m_dictionary );
```



PAIR_ONE2M_FOLDER

```
/* Usage */
pair_one2m_folder_new(
    char *many_folder_name );

char *many_folder_name;

char *pair_folder_element_copy_function(
    prompt_many_element_name,
    many_folder_name,
    pair_one2m_prompt_form_element_name(),
    folder_name );

char *pair_folder_button_string(
    many_folder_name,
    keystrokes_save_function,
    pair_folder_element_copy_function() );
```

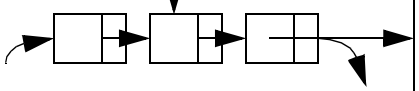
VERTICAL_NEW_BUTTON

```
char *many_folder_name;
char *one_folder_name;

VERTICAL_NEW_BUTTON *vertical_new_button_calloc(
    void );

char *vertical_new_button_one_element_name(
    char *VERTICAL_NEW_BUTTON_ONE_PREFIX,
    char *one_folder_name );

char *vertical_new_button_many_element_name(
    char *VERTICAL_NEW_BUTTON_MANY_LABEL );

APPASERVER_ELEMENT *vertical_new_button_element(
    char *one_folder_name,
    LIST *role_folder_insert_list,
    char *form_name,
    char *VERTICAL_NEW_BUTTON_ONE_PREFIX,
    char *VERTICAL_NEW_BUTTON_DISPLAY,
    char *onclick_keystrokes_save_string );

char *vertical_new_button_dictionary_one_folder_name(
    char *VERTICAL_NEW_BUTTON_ONE_PREFIX,
    DICTIONARY *non_prefixed_dictionary );

char *vertical_new_button_dictionary_folder_name(
    char *hidden_label,
    DICTIONARY *non_prefixed_dictionary );

void vertical_new_button_dictionary_set(
    DICTIONARY *non_prefixed_dictionary,
    char *hidden_label,
    char *folder_name );
```

INSERT

```
char *application_name;
char *session;
char *folder_name;
LIST *primary_key_list;
LIST *attribute_name_list;
LIST *data_list;
DICTIONARY *row_dictionary;
LIST *ignore_insert_attribute_name_list;
boolean insert_row_zero_only;
boolean dont_remove_tmp_file;
int rows_inserted;
char *post_change_process;
char *login_name;
char *tmp_file_name;
FILE *tmp_file;
LIST *attribute_list;
```

```
LIST *attribute_data_list =
    insert_database_attribute_data_list(
        row_dictionary,
        row,
        ignore_insert_attribute_name_list,
        primary_key_list,
        attribute_name_list,
        mto1_related_folder_list,
        common_non_primary_key_list,
        application_name,
        attribute_list );

LIST *insert_attribute_name_list =
    insert_database_attribute_name_list(
        insert_database_attribute_data_list() );
```

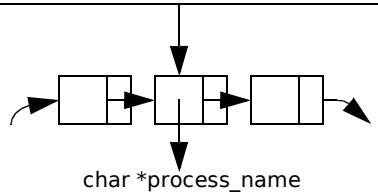


INSERT_DATABASE_ATTRIBUTE_DATA

```
/* Usage */
INSERT_DATABASE_ATTRIBUTE_DATA *
    insert_database_attribute_data_new(
        char *attribute_name,
        char *security_sql_injection_escape(
            security_replace_special_characters(
                string_trim_number_characters(
                    char *data,
                    char *datatype_name ) ) ) );

char *attribute_name;
char *escaped_replaced_data;
```

PROCESS (Generic)

```
boolean process_prompt_exists_preprompt(
    char *process_or_set_name );

JAVASCRIPT *javascript_new(
    char *javascript_filename,
    char *appaserver_parameter_file->
        document_root,
    application_relative_source_directory() );

boolean process_executable_ok(
    char *environment_application_name(),
    char *role_name,
    char *column( command_line, 0 )
        /* executable_filename */ );

char *process_name_fetch(
    char *process_or_set_name );

char *process_set_name_fetch(
    char *process_or_set_name );

LIST *process_role_process_name_list(
    role_primary_where(
        char *role_name ) );
```



PROCESS (Command line)

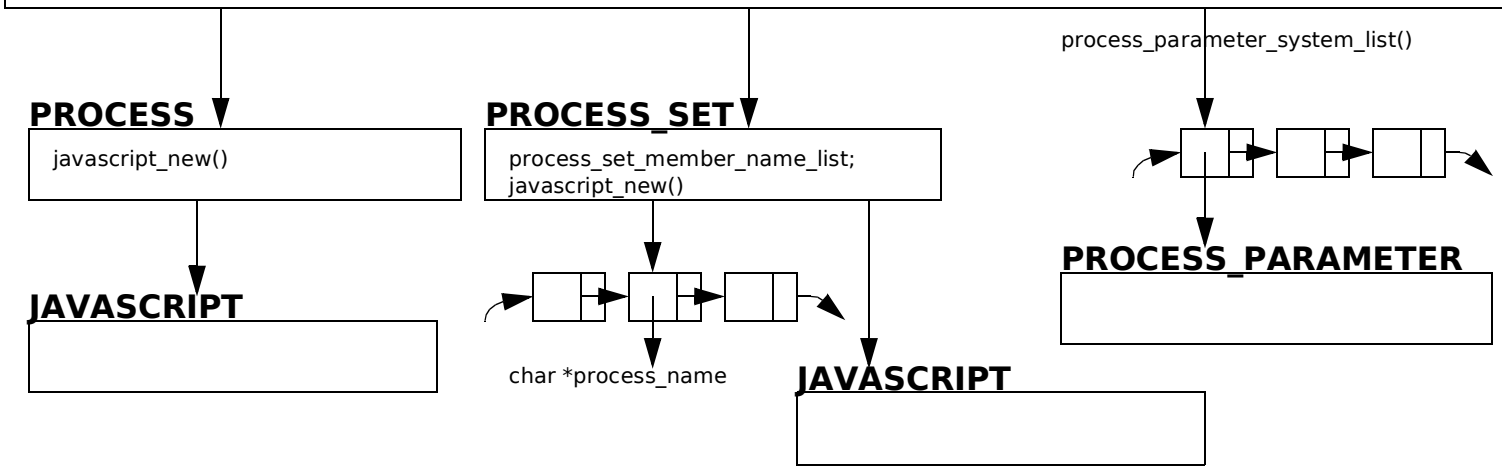
```
char *process_choose_isa_command_line(
    char *command_line /* in */,
    char *application_name,
    char *security_entity_where,
    char *login_name,
    char *role_name,
    char *one2m_isa_folder_name );

char *process_operation_command_line(
    char *command_line /* in */,
    char *environment_application_name(),
    char *operation_name,
    char *login_name,
    char *role_name,
    char *folder_name,
    pid_t parent_process_id,
    char *session_key,
    int operation_row_total(),
    LIST *operation_row_primary_data_list(),
    DICTIONARY *operation_single_row_dictionary() );

char *process_drop_down_command_line(
    char *command_line /* in */,
    char *environment_application_name(),
    char *one2m_folder_name,
    char *state,
    DICTIONARY *drillthru_dictionary,
    DICTIONARY *working_post_dictionary );

char *process_update_row_command_line(
    char *command_line /* in */,
    DICTIONARY *drillthru_dictionary,
    char *login_name,
    char *role_name,
    char *folder_name,
    char *state,
    int row,
    char *process_name,
    char *one2m_folder_name,
    LIST *primary_data_list );

char *process_search_replace_where(
    char *command_line /* in */,
    DICTIONARY *drillthru_dictionary,
    char *login_name,
    char *role_name,
    char *folder_name );

char *process_replace_one2m_folder_name(
    char *command_line /* in */,
    char *one2m_folder_name );
```

PROCESS_PROMPT_OUTPUT

```
/* Usage */
PROCESS_PROMPT_OUTPUT *
    process_prompt_output_fetch(
        char *process_or_set_name,
        char *role_name,
        char *login_name,
        boolean is_preprompt,
        appaserver_parameter_file->
            document_root,
        application_relative_source_directory(),
        DICTIONARY *drillthru_dictionary );

/* Input */
char *process_or_set_name;
char *role_name;
char *login_name;
boolean is_preprompt;
char *document_root_directory;
relative_source_directory;
DICTIONARY *drillthru_dictionary;
```

```
/* Process */
char *process_name =
    process_name_fetch(
        process_or_set_name );

char *process_set_name =
    process_set_name_fetch(
        process_or_set_name );

PROCESS *process =
    process_fetch(
        process_name,
        document_root_directory,
        relative_source_directory,
        0 /* not check_executable_inside */ );

PROCESS_SET *process_set =
    process_set_fetch(
        process_set_name,
        role_name,
        1 /* fetch_member_process_name_list */ );
```

```
LIST *process_parameter_list =
    process_parameter_system_list(
        process_set_parameter_system_string(
            process_set_parameter_where(
                process_set_name,
                is_preprompt ) ),
        role_name,
        login_name,
        drillthru_dictionary );

LIST *process_parameter_list =
    process_parameter_system_list(
        process_parameter_system_string(
            process_parameter_where(
                process_name,
                is_preprompt ) ),
        role_name,
        login_name,
        drillthru_dictionary );
```



APPASERVER Data Structure

PROCESS_PROMPT_SUBMIT

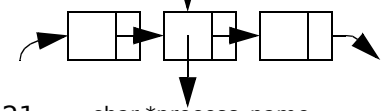
```
/* Usage */
PROCESS_PROMPT_SUBMIT *
    process_prompt_submit_fetch(
        char *process_name,
        char *role_name,
        char *login_name,
        DICTIONARY *working_post_dictionary );

/* Input */
char *process_name;
char *role_name;
char *login_name;
DICTIONARY *working_post_dictionary;

/* Process */
PROCESS *process =
    process_fetch(
        process_name,
        appaserver_parameter_file->
            document_root,
        application_relative_source_directory(),
        1 /* check_executable_inside_filesystem */ );

char *process_prompt_submit_command_line(
    char *command_line,
    char *process_name,
    char *role_name,
    char *login_name,
    DICTIONARY *working_post_dictionary );
```

PROCESS

PROCESS_SET

```
/* Usage */
PROCESS_SET *process_set_fetch(
    char *process_set_name,
    char *role_name,
    char *application_parameter_file->
        document_root,
    char *application_relative_source_directory(),
    boolean fetch_process_set_member_name_list );

/* Attributes */
char *notepad;
char *html_help_file_anchor;
char *post_change_javascript;
char *prompt_display_text;
char *process_group;
char *preprompt_help_text;
boolean prompt_display_bottom;
char *javascript_filename;

PROCESS_SET *process_set_parse(
    char *input,
    char *role_name,
    char *document_root_directory,
    char *application_relative_source_directory,
    boolean fetch_process_set_member_list );

char *process_set_primary_where(
    char *process_set_name );

LIST *process_set_member_name_list(
    process_set_primary_where(),
    role_name );

JAVASCRIPT *javascript =
    javascript_new(
        javascript_filename,
        document_root_directory,
        application_relative_source_directory );

/* Generic */
char *process_set_process_where(
    process_set_member_name_list() );
```

process_set_member_name_list()



char *process_name

PROCESS

```
/* Usage */
PROCESS *process_fetch(
    char *process_name,
    char *application_parameter_file->
        document_root,
    char *application_relative_source_directory(),
    boolean check_executable_inside_filesystem );

/* Input */
char *process_name;
char *document_root_directory;
char *relative_source_directory;

/* Attributes */
char *command_line;
char *notepad;
char *html_help_file_anchor;
int execution_count;
char *post_change_javascript;
char *process_set_display;
char *process_group;
char *preprompt_help_text;
char *javascript_filename;

char *process_primary_where(
    char *process_name );

char *process_system_string(
    char *where );

PROCESS *process_parse(
    char *input,
    char *document_root_directory,
    char *relative_source_directory,
    boolean check_executable_inside_filesystem );

JAVASCRIPT *javascript =
    javascript_new(
        javascript_filename,
        document_root_directory,
        relative_source_directory );
```

JAVASCRIPT

```
/* Usage */
JAVASCRIPT *javascript =
    javascript_new(
        char *javascript_filename,
        appaserver_parameter_file->
            document_root,
        application_relative_source_directory() );

/* Process */
char *javascript_html(
    javascript_filename,
    document_root_directory,
    application_relative_source_directory );

char *javascript_source(
    javascript_html(),
    document_root_directory );

/* Generic */
void javascript_replace_state(
    char *post_change_javascript /* in/out */,
    char *state );

void javascript_replace_row(
    char *post_change_javascript /* in/out */,
    char *row );

char *javascript_post_change_append(
    char *original_post_change_javascript,
    char *post_change_javascript,
    boolean place_first );

void javascript_output_cat(
    FILE *output_stream,
    javascript_source() );

void javascript_output_link(
    FILE *output_stream,
    javascript_html() );
```

PROCESS_PARAMETER

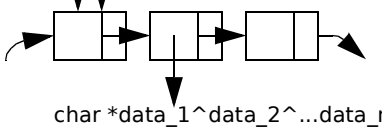
```
/*  Usage */
LIST *process_parameter_system_list(
    char *process_[set_]parameter_system_string(),
    char *role_name,
    char *login_name,
    DICTIONARY *drillthru_dictionary );

/* Input */
char *role_name;
char *login_name;
DICTIONARY *drillthru_dictionary;

/* Attributes */
char *process_or_set_name;
char *folder_name;
char *attribute_name;
char *drop_down_prompt_name;
char *prompt_name;

/* Process */
int display_order;
boolean drop_down_multi_select;
boolean preprompt;
char *populate_drop_down_process_name;
char *populate_helper_process_name;

char *process_parameter_system_string(
    char *where );

char *process_set_parameter_system_string(
    char *where );

PROCESS_PARAMETER *process_parameter_parse(
    char *input,
    char *role_name,
    char *login_name,
    DICTIONARY *drillthru_dictionary );

PROCESS_PARAMETER *process_parameter_new(
    char *process_or_set_name,
    char *folder_name,
    char *attribute_name,
    char *drop_down_prompt_name,
    char *prompt_name );

LIST *process_parameter_process_delimited_list(
    char *folder_name,
    char *populate_drop_down_process_name,
    char *login_name,
    char *role_name,
    DICTIONARY *drillthru_dictionary );

LIST *process_parameter_folder_delimited_list(
    char *widget_folder_name,
    char *login_name,
    char *role_name,
    DICTIONARY *drillthru_dictionary );

ATTRIBUTE *attribute =
    attribute_fetch(
        attribute_name );

DROP_DOWN_PROMPT *drop_down_prompt =
    drop_down_prompt_fetch(
        drop_down_prompt_name );

PROMPT *prompt =
    prompt_fetch(
        prompt_name );

/* Generic */
char *process_parameter_where(
    char *where,
    boolean is_preprompt );
```

ATTRIBUTE

```
char *attribute_name;
```

FOLDER

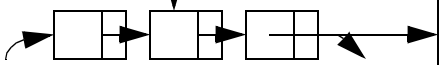
```
char *folder_name;
LIST *primary_delimited_list
LIST *primary_delimited_display_list
```



char *data_1^data_2^...data_n

DROP_DOWN_PROMPT

```
/* Usage */
DROP_DOWN_PROMPT *drop_down_prompt_fetch(
    char *drop_down_prompt_name );

char *drop_down_prompt_name;

/* Process */
char *hint_message;
char *optional_display;

LIST *drop_down_prompt_data_list =
    drop_down_prompt_data_system_list(
        drop_down_prompt_data_system_string(
            drop_down_prompt_primary_where(
                this->drop_down_prompt_name ) ) );
```



DROP_DOWN_PROMPT_DATA

```
char *drop_down_prompt_name;
char *drop_down_prompt_data;
int display_order;
```

PROMPT

```
/* Usage */
PROMPT *prompt_fetch(
        char *prompt_name );

char *prompt_name;

/* Process */
int input_width;
char *hint_message;
boolean upload_filename;
boolean date;
```

PROCESS_GENERIC (1)

```
/* Usage */
PROCESS_GENERIC *process_generic_fetch(
    char *process_set_name,
    char *process_name,
    char *login_name,
    char *output_medium_string,
    DICTIONARY *post_dictionary );

/* Input */
char *process_set_name;
char *login_name;
char *output_medium_string;
DICTIONARY *post_dictionary;

/* Process */
char *process_name =
    process_generic_process_name(
        process_set_name,
        post_dictionary );

char *value_folder_name =
    process_generic_value_folder_name(
        process_name,
        process_set_name );

PROCESS_GENERIC_VALUE_FOLDER *value_folder =
    process_generic_value_folder_fetch(
        value_folder_name );

PROCESS_GENERIC_PARAMETER *parameter =
    process_generic_parameter_parse(
        output_medium_string,
        post_dictionary,
        value_folder->datatype->aggregation_sum,
        value_folder->primary_key_list,
        value_folder->date_attribute_name,
        value_folder->time_attribute_name );

char *process_generic_select(
    value_folder->primary_key_list,
    parameter->value_attribute_name );

char *process_generic_date_where(
    value_folder->date_attribute_name,
    parameter->begin_date,
    parameter->end_date );

PROCESS_GENERIC_WHERE *process_generic_where =
    process_generic_where_new(
        value_folder_name,
        value_folder->foreign_folder->foreign_key_list
            /* where_attribute_name_list */,
        value_folder->foreign_folder->foreign_attribute_data_list
            /* where_attribute_data_list */,
        value_folder->foreign_folder->folder->folder_attribute_primary_list,
        process_generic_date_where() );

char *process_generic_system_string(
    process_generic_select(),
    value_folder_name,
    process_generic_where->where_clause,
    parameter->aggregate_level,
    parameter->aggregate_statistic,
    parameter->accumulate,
    parameter->date_piece,
    parameter->time_piece,
    parameter->value_piece,
    parameter->end_date,
    value_folder->datatype->unit /* datatype_unit */,
    value_folder->foreign_folder->unit /* foreign_folder_unit */,
    parameter->units_converted );
```

PROCESS_GENERIC_VALUE_FOLDER

PROCESS_GENERIC_PARAMETER

PROCESS_GENERIC_WHERE

PROCESS_GENERIC (2)

```
char *process_generic_heading(
    value_folder->primary_key_list,
    value_folder->value_attribute_name,
    value_folder->datatype->unit /* datatype_unit */,
    value_folder->foreign_folder->unit /* foreign_folder_unit */,
    parameter->units_converted,
    parameter->aggregate_level,
    parameter->aggregate_statistic,
    parameter->accumulate );

char *process_generic_subtitle(
    value_folder_name,
    parameter->begin_date,
    parameter->end_date,
    parameter->aggregate_level,
    parameter->aggregate_statistic
    additional_message );

char *process_generic_units_label(
    units,
    units_converted,
    enum aggregate_statistic );

enum date_convert_format process_generic_date_convert_format(
    login_name );

boolean process_generic_no_cycle_color(
    role_name );
```

PROCESS_GENERIC_PARAMETER

```
/* Attributes */
char *output_medium_string;
enum process_generic_medium output_medium;
char *begin_date;
char *end_date;
enum aggregate_level aggregate_level;
enum aggregate_statistic aggregate_statistic;
boolean accumulate;
char *email_address;
char *units_converted;
pidt process_id;
int date_piece;
int time_piece;
int value_piece;

/* Usage */
PROCESS_GENERIC_PARAMETER *process_generic_parameter_parse(
    process_generic->output_medium_string,
    process_generic_>post_dictionary,
    process_generic->value_folder->datatype->aggregation_sum,
    value_folder_primary_key_list,
    value_folder_date_attribute_name,
    value_folder_time_attribute_name );
```

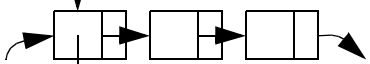
```
{ text_file,
  output_medium_stdout,
  table,
  spreadsheet,
  gracechart,
  output_medium_unknown
```

PROCESS_GENERIC_WHERE

```
/* Usage */
PROCESS_GENERIC_WHERE *process_generic_where_new(
    char *value_folder_name,
    LIST *where_attribute_name_list,
    LIST *where_attribute_data_list,
    LIST *folder_attribute_primary_list,
    char *date_where );

/* Process */
char *data_where =
    query_data_where_clause(
        value_folder_name,
        where_attribute_name_list,
        where_attribute_data_list,
        folder_attribute_primary_list );

char *process_generic_where_clause(
        data_where,
        date_where );
```

PROCESS_GENERIC_VALUE_FOLDER

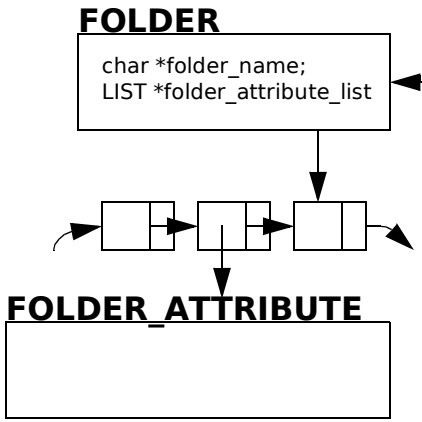
```
/* Usage */
PROCESS_GENERIC_VALUE_FOLDER *value_folder =
    process_generic_value_folder_fetch(
        process_generic->value_folder_name,
        process_generic->post_dictionary );

/* Input */
char *value_folder_name;
DICTIONARY *post_dictionary;

/* Process */
PROCESS_GENERIC_VALUE_FOLDER *value_folder_parse(
    input );

char *datatype_folder_name;
char *foreign_folder_name;
char *datatype_attribute_name;
char *date_attribute_name;
char *time_attribute_name;
char *value_attribute_name;
boolean exists_aggregation_sum;
boolean exists_bar_graph;
boolean exists_scale_graph_zero;
boolean datatype_exists_unit;
boolean foreign_exists_unit;

LIST *folder_attribute_list =
    folder_attribute_fetch_list(
        value_folder_name,
        (LIST *)0 /* exclude_attribute_name_list */ );

PROCESS_GENERIC_DATATYPE *datatype =
    process_generic_datatype_fetch(
        datatype_folder_name,
        datatype_attribute_name,
        exists_aggregation_sum,
        exists_bar_graph,
        exists_scale_graph_zero,
        datatype_exists_units,
        post_dictionary );

PROCESS_GENERIC_FOREIGN_FOLDER *foreign_folder =
    process_generic_foreign_folder_fetch(
        foreign_folder_name,
        foreign_exists_unit,
        post_dictionary );

LIST *process_generic_datatype_compare_list(
    process_generic->post_dictionary,
    exists_aggregation_sum,
    exists_bar_graph,
    exists_scale_graph_zero,
    datatype_exists_unit );
```



char *

PROCESS_GENERIC_DATATYPE

PROCESS_GENERIC_FOREIGN_FOLDER



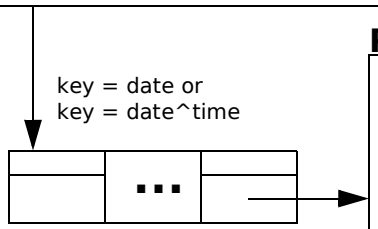
PROCESS_GENERIC_FOREIGN_FOLDER

```
/* Usage */
PROCESS_GENERIC_FOREIGN_FOLDER *process_generic_foreign_folder_fetch(
    process_generic->value_folder->foreign_folder_name,
    process_generic->post_dictionary,
    process_generic->value_folder->foreign_exists_unit );

/* Process */
FOLDER *folder =
    folder_fetch(
        foreign_folder_name,
            (char *)0 /* role_name */,
            (LIST *)0 /* exclude_attribute_name_list */,
            /* Also sets folder_attribute_primary_list */
            /* and primary_key_list. */
            1 /* fetch_folder_attribute_list */,
            0 /* not fetch_relation_mto1_non_isa_list */,
            0 /* not fetch_relation_mto1_isa_list */,
            0 /* not fetch_relation_one2m_list */,
            0 /* not fetch_relation_one2m_recursive_list */,
            0 /* not fetch_process */,
            0 /* not fetch_role_folder_list */,
            0 /* not fetch_row_level_restriction */,
            0 /* not fetch_role_operation_list */ );

LIST *foreign_key_list = folder->primary_key_list;

LIST *foreign_attribute_data_list =
    dictionary_key_list_fetch(
        foreign_key_list,
        post_dictionary );

char *unit = (foreign_exists_unit) ? list_last( foreign_attribute_data_list ) : (char *)0;
```

PROCESS_GENERIC_DATATYPE

```
/* Usage */
PROCESS_GENERIC_DATATYPE *process_generic_datatype_fetch(
    process_generic_value_folder->datatype_folder_name,
    process_generic_value_folder->datatype_attribute_name,
    process_generic_value_folder->exists_aggregation_sum,
    process_generic_value_folder->exists_bar_graph,
    process_generic_value_folder->exists_scale_graph_zero,
    process_generic_value_folder->datatype_exists_units,
    process_generic_value_folder->post_dictionary );

/* Input */
char *datatype_folder_name;
char *datatype_attribute_name;
boolean exists_aggregation_sum;
boolean exists_bar_graph;
boolean exists_scale_graph_zero;
boolean datatype_exists_units;
DICTIONARY *post_dictionary;

/* Process */
char *datatype_name;
boolean aggregation_sum;
boolean bar_graph;
boolean scale_graph_zero;
char *unit;
```

```
char *process_generic_datatype_select(
    datatype_attribute_name,
    exists_aggregation_sum,
    exists_bar_graph,
    exists_scale_graph_zero,
    datatype_exists_unit );

char *datatype_name =
    dictionary_get(
        datatype_attribute_name,
        post_dictionary );

char *process_generic_datatype_where(
    datatype_attribute_name,
    datatype_name );

char *process_generic_datatype_system_string(
    process_generic_datatype_select(),
    process_generic_datatype->datatype_folder_name,
    process_generic_datatype_where() );

PROCESS_GENERIC_DATATYPE *process_generic_datatype_parse(
    string_pipe_fetch(
        process_generic_datatype_system_string() )
            /* input_buffer */ );
```



PROCESS_GENERIC_VALUE

```
char *date;
char *time;
char *unit;
double value;
boolean is_null;
double accumulate
```

```
PROCESS_GENERIC_VALUE *process_generic_value_parse(
    double *accumulate_value,
    char *input,
    char *time_attribute,
    boolean unit_value_folder,
    boolean accumulate_boolean,
    enum aggregate_level );
```

POST_DICTIONARY

```
/* Usage */
POST_DICTIONARY *post_dictionary_stdin_new(
    /* Used when expecting a spooled file */
    char *appaserver_data_directory,
    char *session_key );

POST_DICTIONARY *post_dictionary_string_new(
    char *dictionary_string );

/* Process */
char *post_dictionary_apache_key(
    string_input() );

DICTIONARY *original_post_dictionary =
    post_dictionary_fetch(
        FILE *stdin,
        char *appaserver_data_directory,
        char *session_key,
        post_dictionary_apache_key() );

char *post_dictionary_apache_label(
    char *input );

char *post_dictionary_separate_key(
    char *input );

char *post_dictionary_input_filename(
    char *input );

void post_dictionary_stream_set(
    DICTIONARY *original_post_dictionary /* out */,
    FILE *stdin,
    post_dictionary_separate_key(),
    post_dictionary_apache_key() );

char *post_dictionary_spool_filename(
    search_replace_character(
        post_dictionary_input_filename(),
        ' ',
        '_' ),
    appaserver_data_directory,
    session_key );

void post_dictionary_spool_file(
    FILE *stdin,
    char *apache_key,
    post_dictionary_spool_filename() );
```

DICTIONARY_SEPARATE

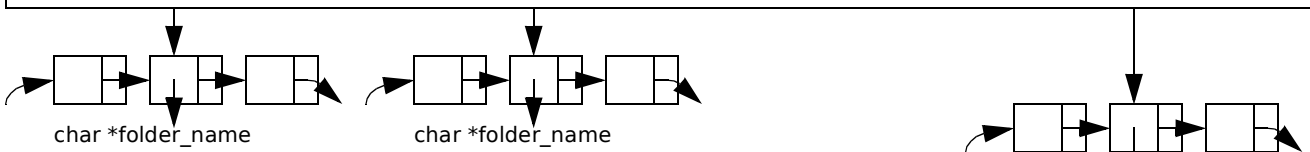
```
/* Usage */
DICTIONARY_SEPARATE *dictionary_separate_folder_new(
    DICTIONARY *original_post_dictionary,
    char *application_name,
    char *login_name,
    LIST *folder_attribute_date_name_list(
        folder->folder_attribute_list ) );

DICTIONARY_SEPARATE *dictionary_separate_row_new(
    DICTIONARY *post_dictionary_row_new()->row_dictionary
        /* original_row_dictionary */,
    char *application_name,
    char *login_name,
    LIST *folder_attribute_name_list(
        folder->folder_attribute_list ),
    LIST *folder_attribute_date_name_list(
        folder->folder_attribute_list ),
    LIST *operation_name_list );

DICTIONARY_SEPARATE *dictionary_separate_string_new(
    DICTIONARY *post_dictionary_string_new()->dictionary
        /* original_post_dictionary */ );

/* Process */
DICTIONARY *dictionary_separate_working_post(
    DICTIONARY *original_post_dictionary,
    char *application_name,
    LIST *folder_attribute_date_name_list,
    char *login_name );

DICTIONARY *dictionary_separate_sort(
    dictionary_separate_working_post() );

DICTIONARY *dictionary_separate_query(
    dictionary_separate_working_post() );

DICTIONARY *dictionary_separate_drillthru(
    dictionary_separate_working_post() );

DICTIONARY *dictionary_separate_ignore(
    dictionary_separate_working_post() );

LIST *dictionary_separate_ignore_select_attribute_name_list(
    dictionary_separate_ignore() );

DICTIONARY *dictionary_separate_pair_one2m(
    dictionary_separate_working_post() );

DICTIONARY *dictionary_separate_non_prefixed(
    dictionary_separate_working_post() );

DICTIONARY *dictionary_separate_row_zero(
    dictionary_separate_working_post(),
    folder_attribute_name_list );

void dictionary_separate_date_convert(
    DICTIONARY *working_post_dictionary,
    char *application_name,
    LIST *folder_attribute_date_name_list,
    char *login_name );

void dictionary_separate_string_date_convert(
    DICTIONARY *dictionary,
    char *application_namem,
    char *date_string,
    char *key,
    char *login_name );

DICTIONARY *dictionary_separate_row_dictionary_multi_row(
    dictionary_separate_non_prefixed(),
    folder_attribute_name_list,
    operation_name_list );

void dictionary_separate_parse_multi_attribute_keys(
    DICTIONARY *working_post_dictionary,
    char *prefix );

void dictionary_separate_trim_multi_drop_down_index(
    DICTIONARY *working_post_dictionary,
    char multi_select_move_left_right_index_delimiter );

void dictionary_separate_trim_double_bracked_data(
    DICTIONARY *working_post_dictionary );

/* Generic */
DICTIONARY *dictionary_separate_send_dictionary(
    DICTIONARY *sort_dictionary,
    DICTIONARY *query_dictionary,
    DICTIONARY *drillthru_dictionary,
    DICTIONARY *ignore_dictionary,
    DICTIONARY *pair_one2m_dictionary,
    DICTIONARY *non_prefixed_dictionary );

char *dictionary_separate_send_string(
    dictionary_separate_send_dictionary() );
```

FOLDER_MENU

```
/* Usage */
FOLDER_MENU *folder_menu_new(
    char *application_name,
    char *session_key,
    char *appaserver_data_directory,
    char *role_name );

/* Input */
char *application_name;
char *session_key;
char *appaserver_data_directory;
char *role_name;
```

```
/* Process */
ROLE *role =
    role_fetch(
        role_name,
        0 /* not fetch_role_attribute_exclude_list */ );

LIST *folder_menu_lookup_folder_name_list(
    char *role_name );

LIST *folder_menu_insert_folder_name_list(
    char *role_name );

if ( role->folder_count )
```

```
{
    char *folder_menu_filename(
        char *application_name,
        char *session_key,
        char *appaserver_data_directory,
        char *role_name );

    LIST *folder_menu_lookup_count_list(
        application_name,
        folder_menu_lookup_folder_name_list(),
        folder_menu_filename() );
}
```



FOLDER_MENU_COUNT

```
/* Usage */
LIST *folder_menu_lookup_count_list(
    char *application_name,
    folder_menu_lookup_folder_name_list(),
    folder_menu_filename() );

/* Input */
char *application_name;
LIST *lookup_folder_name_list;
char *filename;
```

```
/* Attributes */
char *folder_name;
long int count;

FOLDER_MENU_COUNT *folder_menu_count_fetch(
        char *application_name,
        char *folder_name );

/* Process */
```

```
LIST *folder_menu_read_count_list(
    char *filename );

long int folder_menu_count(
    char *folder_table_name() );

void folder_menu_write_count_list(
    char *filename,
    folder_menu_lookup_count_list() );
```

SESSION

```
/* Usage */
/* Sets appaserver environment and outputs argv */
/* Each parameter is security inspected */
SESSION *session_folder_integrity_exit(
    int argc,
    char **argv,
    char *application_name,
    char *login_name,
    char *session_key,
    char *folder_name,
    char *role_name,
    char *state );

/* Sets appaserver environment and outputs argv */
/* Each parameter is security inspected */
SESSION *session_process_integrity_exit(
    int argc,
    char **argv,
    char *application_name,
    char *login_name,
    char *session_key,
    char *process_name,
    char *role_name );

SESSION *session_fetch(
    /* Sets APPASERVER_DATABASE */
    char *security_sql_injection_escape(
        application_name ),
    char *security_sql_injection_escape(
        session_key ),
    char *login_name
        /* integrity_check_login_name */ );

LIST *session_system_list(
    session_system_string() );

/* Input */
char *sql_injection_escape_application_name;
char *sql_injection_escape_session_key;

/* Attributes */
char *session_key;
char *login_name;
char *login_date;
char *login_time;
char *last_access_date;
char *last_access_time;
char *http_user_agent;
char *remote_ip_address;

/* Process */
char *sql_injection_escape_role_name =
        security_sql_injection_escape(
            role_name );

char *sql_injection_escape_folder_name =
        security_sql_injection_escape(
            folder_name );

char *session_current_ip_address(
    void );

char *session_primary_where(
    char *session_key );

char *session_system_string(
    char *where );

boolean session_remote_ip_address_changed(
    session->remote_ip_address,
    session_currrent_ip_address() );

void session_message_ip_address_changed_exit(
    char *sql_injection_escape_application_name,
    char *session_key,
    char *remote_ip_address,
    char *session_current_ip_address(),
    char *login_name );

void session_access_failed_message_exit(
    char *application_name,
    char *session_current_ip_address(),
    char *login_name );

char *session_state_integrity(
    char *state,
    LIST *role_folder_fetch_list(
        sql_injection_escape_role_name,
        sql_injection_escape_folder_name ) );

void session_update_access_date_time(
    char *session_key );

void session_environment_set(
    char *application_name );
```

POST_LOGIN

```
/* Usage */
POST_LOGIN *
    prompt_edit_form_new(
        int argc,
        char **argv );

/* Process */
DICTIONARY *post_login_dictionary( void );

char *post_login_application_name(
    int argc,
    char **argv,
    post_login_dictionary() );

char *sql_injection_escape_login_name =
    security_sql_injection_escape(
        dictionary_get(
            “login_name”,
            post_login_dictionary() );

char *sql_injection_escape_password =
    security_sql_injection_escape(
        dictionary_get(
            “password”,
            post_login_dictionary() );

boolean post_login_missing_name(
    sql_injection_escape_login_name );

boolean post_login_public_name(
    sql_injection_escape_login_name );

char *post_login_database_password(
    post_login_application_name(),
    sql_injection_escape_login_name );

boolean post_login_missing_database_password(
    sql_injection_escape_password );

boolean post_login_name_email_address(
    sql_injection_esacpe_login_name );

enum post_login_password_match_return
    post_login_password_match(
        post_login_application_name(),
        post_login_name_email_address(),
        post_login_public_name(),
        sql_injection_escape_password,
        post_login_database_password() );

char *post_login_session(
    post_login_application_name(),
    sql_injection_escape_login_name );

void post_login_frameset_output(
    post_login_application_name(),
    sql_injection_escape_login_name,
    post_login_session() );

enum post_login_password_match_return
```

```
password_match,
password_fail,
public_login,
email_login
```